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Why should my group trust yours? Collective trust and trustworthiness under Economic Shocks

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Universidad de los Andes Colombia | Facultad de Economía

Why should my group trust yours?
Collective trust and trustworthiness under Economic Shocks*

Andres Barinas-Forero†

Abstract

While trust remains pivotal for economic growth in any society, its consistency over time is not guaranteed. External shocks such as economic crises or natural disasters can disrupt group trust dynamics. Through a laboratory experiment, this study examines the effects of random negative economic shocks on economic relations characterized by having a group of trustors and a group of trustees. The results indicate that such shocks decrease collective trust between groups due to two main factors: firstly, a decline in the trustor's perceived trustworthiness of trustees; and secondly, a tendency for trustors to rely on the contributions of other trustors, leading to free-riding behavior. Furthermore, the experiment reveals that groups become less trustworthy following the shock, as trustees prioritize recovering economic gains from trust relations. This decline in collective trustworthiness is driven by diminished trustee's beliefs in the trustworthiness of other trustees and trustees' enforcing what they consider socially appropriate to do: return less after the shock.

Keywords: Trust, Trustworthiness, Collective Trust, Negative Shocks, Groups, Laboratory Experiments
JEL Classification: C92, C91, D91, D31

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¿Por qué mi grupo debería confiar en el tuyo?
Confianza y confiabilidad colectiva bajo choques económicos*

Andres Barinas-Forero†

Abstract

Aunque la confianza sigue siendo fundamental para el crecimiento económico, su consistencia a lo largo del tiempo no está garantizada. Choques externos, como las crisis económicas o los desastres naturales, pueden alterar la dinámica de confianza grupal. Empleando un experimento de laboratorio, este artículo examina los efectos de choques económicos negativos aleatorios sobre relaciones económicas caracterizadas por la presencia de un grupo de confiadores y un grupo de receptores de confianza (fiduciarios). Los resultados indican que dichos choques disminuyen la confianza colectiva entre grupos debido a dos factores: en primer lugar, una disminución en la percepción de confiabilidad de los fiduciarios por parte de los confiadores; en segundo lugar, una tendencia de los confiadores a depender de las contribuciones de otros confiadores, lo que conduce a comportamientos de aprovechamiento gratuito. Además, los resultados revelan que los grupos se vuelven menos confiables tras el choque, ya que los fiduciarios priorizan la recuperación de ganancias económicas propias de relaciones de confianza. Esta disminución en la confiabilidad colectiva se debe a la disminución de las creencias de los fiduciarios en la confiabilidad de los demás y a que los fiduciarios imponen lo que consideran socialmente apropiado: devolver menos después del choque.

Palabras clave: Confianza, Confiabilidad, Confianza colectiva, Choques económicos, Grupos, Experimento de laboratorio
Clasificación JEL: C92, C91, D91, D31

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1 Introduction

Trust is commonly regarded as one of the critical factors of economic growth and development given the benefits it brings to social, economic, and commercial relationships (Arrow, 1972; Putnam, 1993; Whiteley, 2000; Dearmon and Grier, 2009; Algan and Cahuc, 2013). In societies with high levels of interpersonal trust, civic cooperation is higher (Knack and Keefer, 1997), transaction costs are lower (Dyer and Chu, 2003), and investment is more abundant (Zak and Knack, 2001). Overall, higher levels of trust and trustworthiness make social interactions more efficient due to lower beliefs of opportunistic behavior.

Given the pivotal role of trust in fostering economic development, investigating its determinants and volatility remains crucial in the field of economics. Although several authors have highlighted the importance of contextual influences (Glaeser et al., 2000), emotional factors (Dunn and Schweitzer, 2005), economic conditions (LaPorta et al., 1997), or traumatic events (Schwerter and Zimmermann, 2020; Alesina and La Ferrara, 2002) over economic trust, there is also a growing body of literature that shows the harmful impacts that economic shocks have over interpersonal trust (Fleming et al., 2014; Bejarano et al., 2018; Azzollini, 2023; Frost et al., 2024), making relevant the study of *trust dynamics* after negative economic shocks.

While some economic interactions rely on interpersonal trust, many others depend on group trust relationships (Bachmann and Zaheer, 2006; Charness and Sutter, 2012; McEvily et al., 2002) that are also affected by economic shocks. These situations are characterized by having a group of trustors and a group of trustees (Kugler et al., 2007; Holm and Nystedt, 2010) that interact in relations of **collective trust** and **collective trustworthiness**. To exemplify this, consider a local manager inside a traditional restaurant that establishes the following tips policy: *"All tips are collected by waiters and are equally distributed among waiters and cooks at the end of the night"*. Here we have a **collective trust** relationship where cooks (group of *trustors*) trust that waiters (group of *trustees*) will report the total amount of tips at the end of the day, and waiters decide to be trustworthy by not keeping tips for themselves. Moreover, this constitutes a collective trust relationship because a specific cook shows his trust to the entire group of waiters, and a specific waiter shows his trustworthiness to the entire group of cooks.[1]

For instance, consider an economic crisis that motivates the restaurant manager to implement salary cuts for the group of waiters. Consequently, the manager informs all restaurant employees (both cooks and waiters) about the imminent reduction in the waiter's wages. Given that this

[1] Despite this seeming to be a particular example, collective trust and trustworthiness relationships occur in multiple dimensions beyond this initial framework. For example, in politics, citizens trust politicians within congress, and congress members work as a group to seem trustworthy. In education, parents of students trust teachers to provide quality education for their children within a given school. Just like in the restaurant, legislative branch, or traditional school, some scenarios are characterized by the impossibility of establishing interpersonal trust relationships.

economic shock impacts the entire group of waiters, it may be ambiguous how this event influences the trust dynamics between the two groups. This ambiguity arises because trust is no longer solely based on individual preferences, personal intrinsic reward for trust (Andreoni, 1990) or beliefs about the behavior of members of the opposite group (Ashraf et al., 2006; Kim et al., 2022), but also on additional factors like beliefs about the behavior of members of my group, which may result in trust mediated by conditional reciprocity (Fischbacher et al., 2001). Indeed, as a waiter, the belief that my fellow waiters will take the tips for themselves makes it more costly for me to be trustworthy since they are free-riding on my tips declaration (Fischbacher and Gächter, 2010), and as a cook, a lower belief in the trust placed by my fellow's cooks may incentivize me to reduce my trust and my cooking effort (since one well-done dish will not make a difference on total tips).

Therefore, it is clear that understanding how economic shocks affect collective trust and trustworthiness is particularly difficult due to the individual and group incentives at play during group trust interactions. Contrary to scenarios of interpersonal trust, it is unclear how cooks (*trustors*) will respond to a decrease in waiters' (*trustees'*) income, given that *some* cooks may help waiters with the crisis (Fehrler, 2010), while others may mistrust waiters and opt to reduce their trust. This ambiguity leaves us with an open question: What are the effects of negative economic shocks on collective trust and collective trustworthiness?

In this paper, I develop a theoretical model of collective trust, showing how random negative economic shocks affect collective trust and collective trustworthiness. I focus on the effect of an exogenous shock to trustees' income that can exemplify an unemployment crisis (Azzollini, 2023), a wage income reduction (Ananyev and Guriev, 2019), unexpected destruction of wealth, or physical capital destruction (You et al., 2020; Bai and Li, 2021; Frost et al., 2024). I begin presenting an *homoeconomicus* rationale that considers agents with perfect rationality and self-interested goals. I find that trustors and trustees do not have incentives to change their behaviors after the shock.[2] Subsequently, I relax the self-interested agent's assumption by considering a trustor that cares about trustees payoff (Fehr and Schmidt, 1999; Bolton and Ockenfels, 2000). In particular, I assume that the trustor is an *inequality-averse* agent that does not have a preference for a particular trustee but instead cares for the whole group of trustees. I refer to this setting as the *homoaltruistic* rationale, which consistently captures other equilibriums that differ from the zero trust level prediction of the *homoeconomicus* model. (Henrich et al., 2001; Cox, 2004; Fehr, 2009; Johnson and Mislin, 2011; Aĺos-Ferrer and Farlofi, 2019).

The *homoaltruistic* rationale suggest that a negative economic shock should increase the trust placed in the group of trustees. Moreover, the increase in collective trust could be higher when the inequality-averse sender believes that others will cooperate more after the shock. Conversely,

[2] Furthermore, the homoeconomicus model predicts that trustees and trustors do not have a collective action problem at all; hence, we would not expect differences between interpersonal trustworthiness and collective trustworthiness

multiple factors influence and modify trustee's trustworthiness after the shock, although it is hard to tell exactly how those factors can impact the outcome. First, the negative economic shock could reduce trustworthiness if the trustee uses the funds to overcome the crisis instead of making repayment arrangements (Bejarano et al., 2021), and this reduction could be even more salient and pronounced if the trustee hypothesizes that his fellows will do the same. Alternatively, trustworthiness could increase after the shock if trustees place more value on the signal of being trustworthy under unfavorable conditions or if they value more the trust placed by trustors under a higher opportunistic context, given the shock.

In order to provide an empirical answer to the research question, I implement a modified trust game experiment (Berg et al., 1995) under a collective (group) environment (Holm and Nystedt, 2010) with 476 college students in Bogotá, Colombia. For the experimental design, I employed a between-subjects design randomizing the occurrence of a negative economic shock that destroys **60%**[3] **of the trustee's** endowment. This allows me to compare the collective trust and trustworthiness under unexpected income reductions. In particular, participants engage in a two-round experiment, being the first round a collective trust interaction with equal endowments for all participants and treatment groups. In the second round, the negative shock occurs only for the treatment group, while the control group remains with the previous setting. A first non-shocked round allows me to compare the "pre-treatment" behavior of collective trust between both treatments. Also, it makes the inequality generated by the shock more salient. It is worth mentioning that the occurrence of the shock is common knowledge, so both trustors (henceforth Senders) and trustees (henceforth Receivers) know about it, and in particular, both agents know the magnitude of the shock. I follow standard procedures involving trust games, so I consider trust to be measured as the *amount sent* (as a percentage of the initial endowment) by each sender and trustworthiness as the *amount returned* (as a percentage of the amount received) by each receiver. Further details of the experimental design are presented in Section 3.

In a general framework, and regarding collective trust behavior in the first round of the game, I find that collective trust is close to 43% (SD = 0.28), and statistically different from zero. Given that this measure is captured at the individual level, it is possible to compare this collective trust measure with the interpersonal trust measure. Overall, I find that collective trust is not statistically different than interpersonal trust where the amount sent is 41% (p-value = 0.11) being this behavior consistent with measures of trust in the region (Cardenas et al., 2013; Ginzburg and Guerra, 2023). Additionally, regarding collective trustworthiness, I find that receiver's average amount returned in the round before the shock is close to 29% (SD = 0.18), and once more is statistically different from zero. Moreover, I find that collective trustworthiness is lower than interpersonal trustworthiness, where the amount returned is 31% (SD = 0.17).

[3] Shock's size is standard considering Bejarano et al. (2021); Aycinena and Blanco (2022); Bogliacino et al. (2022)

With respect to the impact of a negative income shock over trust, I find that the occurrence of the shock reduces collective trust and collective trustworthiness. Furthermore, these results appear to be driven by (1) changes in the first-order beliefs about the behavior of members of the opposite group (Ashraf et al., 2006), (2) changes in the empirical expectations about the behavior of members of my own group, and (3) changes in the perception of what is *socially appropriate* to do after the shock (social norms). Economic shocks significantly impact the dynamics of trust and cooperation in social structures. These shocks have far-reaching effects on society that go beyond simple monetary changes. First of all, changes in the structure of the economy modify the incentives that have been set up for collaboration in previous interactions, which frequently leads to a deterioration in trust levels. Second, according to these findings, participants' beliefs about other's behavior (his fellows and members from the opposite group) noticeably deteriorate after economic shocks. Hence, this quick erosion in trust shows us that, under collective interactions, the trust equilibrium appears to be more sensible. Overall, point estimates demonstrate significant economic implications, indicating that negative economic shocks result in a reduction of trust ranging from -19% to -12% compared to the control group mean, and a reduction in collective trustworthiness of approximately -29% compared to the control group. These results are robust to the inclusion of sociodemographic controls.

Some papers are related to this. For instance, Holm and Nystedt (2010) incorporate a collective mechanism decision in the trust process focusing on estimating the main differences between interpersonal trust and collective trust. I deviate from this study by allowing a non-binary collective trust choice, which provides heterogeneous reaction against shocks, and I introduce a random negative shock to evaluate his impacts. Other papers aim to measure how collective trust relationships differ from interpersonal trust relationships by comparing groups against one individual (McEvily et al., 2002) or between individuals and one group (Currall and Inkpen, 2003), finding that interpersonal trust may differ from trust in a collective environment, being lower the latter (Kugler et al., 2007; Cox, 2002). Although these studies introduce group dynamics of trust, each group conducted face-to-face discussions within the group to decide on a unique strategy. I contribute to the literature by allowing each participant within the group to make his own private choice, so differences once the shock occurs may appear at the individual level, and not just at the group level.

Regarding effects of economic shocks over trust, Bejarano et al. (2018) investigates whether interpersonal trust-related behavior is affected by inequality all by itself or inequality brought on by a negative shock on wealth for trustees. Their findings suggest that a negative economic shock erodes trust and also reduces the trustworthiness. However, little is known about the effects mediated by changes in first-order beliefs. On the contrary, Aycinena and Blanco (2022) examine the impact of exposure to different sets of economic shocks that destroy trustor's and/or trustee's endowments. Their results suggest that exposure to negative economic shocks reduces trust, while being affected by economic shocks increases the amount sent. These results, as explained by Aycinena and

Blanco (2022), can be attributed to changes in the sender's perceptions of trustworthiness. I deviate from this study by allowing all participants to be informed about the realization of the shock and allowing participants to interact in a collective environment. Overall, there is general consensus regarding the detrimental effects of economic shocks on trustee's endowments over interpersonal trust (Bejarano et al., 2021; Bogliacino et al., 2022). Conversely, concerning collective trust, there exists a lack of consensus regarding the impacts of these types of shocks. While some studies suggest that group trust may increase following economic shocks (Navarro-Carrillo et al., 2018), others suggest that trust might be lower given unethical behavior from groups (Kocher et al., 2018) so, overall, little is known about how groups react to those random shocks. This paper aims to fill this empirical gap.

The rest of the paper is organized in 6 sections. In Section 2, I set up a simple model that describes collective trust and trustworthiness under a negative economic shock. In Section 3, I present in detail the experimental design. In Section 4, I present the main findings of this paper. In Section 5, I present empirical evidence of the proposed mechanisms behind these results and Section 6 concludes.

2 Theoretical Model

Consider the following dynamic game with two types of agents: **senders** (*trustors*) and **receivers** (*trustees*). In this set up, the society has $i = \{1, 2, ..., S\}$ number of **senders** and $j = \{1, 2, ..., R\}$ number of **receivers** that interact in a one-shot collective game. Both types of agents start with the same endowment $E^s = E^r$ where both $E^{i,j} \in \mathbb{R}^+$. In particular, E^s denotes the initial endowment of each sender, and E^r denotes the initial endowment of each receiver. In this society, the initial endowment represents assets, salary, or wealth in general. This means that both types of agents have the same wealth, and there are no economic inequalities before the interaction. Both agents know that their initial endowment is the same and that their role is assigned randomly, avoiding generating a sense of belonging among the groups. The timing of the model goes as follows:

Collective Trust Stage

In the first stage of the game, each sender decides how much of his endowment (E_i^s) he wants to transfer to receivers by choosing $M_i^s \in [0, E_i^s]$. All senders decide privately and simultaneously over M_I^s and once all decide, the funds are totaled up ($\sum_{i=1}^{S} M_i$), multiplied by κ (with $\kappa > 1$), and they are sent to the receivers in equal parts. In this sense, each receiver obtains $\left(\frac{\kappa}{R} \sum_{i=1}^{S} M_i^s\right)$.

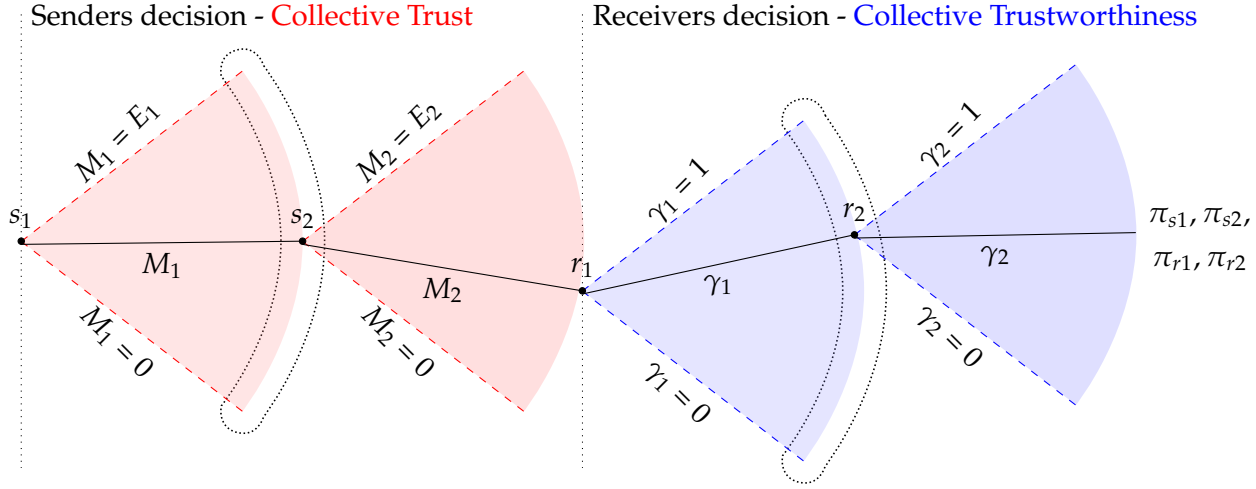
Collective Trustworthiness Stage

In the second stage of the game, the receiver observes the amount transferred to him. Then, he decides how much of the transferred amount he wants to return to the senders by choosing a fraction $\gamma_j^r \in [0,1]$ of his total transfer. Once again, each receiver decides privately and simultaneously over γ^r and once all decide, the returned funds are totaled up $\left(\sum_{j=1}^{R} \gamma_j^r \left(\frac{\kappa}{R} \sum_{i=1}^{S} M_i^s\right)\right)$ and they are sent back to the group of senders in equal parts.

It is worth mentioning that all players have information about the structure of the game, so each sender knows that his contributions are totalled up with contributions of others, while each receiver knows that his amount returned is totalled up with the amount returned by others. Without loss of generality, I can represent the model described above for a finite number of players. For instance, let us consider a 4-players situation with two senders ($S = 2$) and two receivers ($R = 2$) that allows us to see, graphically, the extensive representation of the game (Figure 1).

From Figure 1, observe that the game is finite, dynamic, and has perfect information. It has two different sub-games starting at node 1 and 3, respectively, that do not depend on the number of agents. In the first sub-game, the senders simultaneously decide how much to transfer to receivers. In the second sub-game, the receivers know precisely how much they received (complete information), and they decide simultaneously how much to return. Moreover, note that even though the interaction between groups is dynamic, the decision within each group is static. In this sense, each type of player must enforce his best response in front of his fellow players of the same type.

Figure 1: Extensive representation of the model.



Note: Sender's decision is presented in red. Notice that each sender has a continuous action set between 0 and his total endowment (E_i), and the following sender has only one information set across all possible transfers of his partner (simultaneous game). Receiver's decision is presented in blue. It is worth mentioning that receivers observe the **total** transferred amount as a group, insted of knowing each sender's decision. Once more, receiver's decisions are simultaneous (denoted in one unique information set for each node).

2.1 *Homo Economicus* Rationale

Assuming perfectly rational individuals with ideal judgment for decision-making, with self-interested goals and with no interdependent or other regarding preferences, I characterize a *homo economicus* equilibrium, where each player optimizes their material payoff. Given that the game is finite, dynamic, and has perfect information, it is possible to employ backward induction to solve for the sub-game perfect equilibrium (Selten, 1965).

In the latter stage of the interaction, under the assumption of perfect rationality and perfect recall, receivers solve their maximization problem by optimizing:[4]

$$\max_{\gamma_j^r} \pi_j^r(\gamma_j^r) = E_j^r + \left(\frac{\kappa}{R}\sum_{i=1}^{S} M_i^s\right)(1-\gamma_j^r), \tag{1}$$

where M_i^s denotes the sent amount by sender i, E_j^r denotes the receiver's j endowment, κ is the multiplication factor, S and R denote the total number of senders and receivers respectively and γ_j^r denotes the fraction that receiver j wants to return. To solve for an equilibrium, each receiver maximizes his material payoff function by choosing γ_j^r. As can be seen, any strict positive value for γ_j^r reduces the receiver's payoff. Hence, his optimal solution for this problem is to choose $\gamma_j^r = 0$. The problem is symmetric for all players; for that reason, the Nash equilibrium for this sub-game is found when every receiver plays his best response, leading to a $\gamma_j^r = 0 \ \ \forall j$ strategy.

Note that under this framework, receivers have no collective action problems between them. As shown in Equation (1), the receiver only focuses on how much of his earnings he wants to return, but there is no focus on how much other receivers return. Moreover, there is no relevance either in the amount sent by senders in the first place, as the receiver takes that amount as given. According to the *homo economicus* rationale, choosing to be trustworthy should not be influenced by my beliefs on how trustworthy others are. This is shown in the Equation (1), where it is clear that the decision of γ_j^r is independent of how much I think my fellows trustees will return.

Given the structure of the game, in the first stage of the interaction each sender solve his maximization problem by optimizing the following material payoff:[5]

$$\max_{M_i^s} \pi_i^s = E_i^s + \frac{1}{S}\left[\frac{\kappa}{R}\left(\sum_{i=1}^{S} M_i^s\right)\left(\sum_{j=1}^{R}\gamma_j^r\right)\right] - M_i^s. \tag{2}$$

Foreseeing the strategic behavior of the receivers described above, the sender assumes $\gamma_j^r = 0$

[4] See Appendix A.2 for more details in the construction of the Receiver's payoff
[5] Similarly, details on the construction of the Sender's payoff are presented in Appendix A.1.

and then he maximizes his payoff by choosing M_i^s. As can be seen, any strict positive value for M_i^s reduces the sender's payoff. Consequently, his optimal solution is to play $M_i^s = 0$, and given that the problem is symmetric for all senders, the Nash equilibrium for this sub-game is found when every sender plays $M_i^s = 0$. Thus, the sub-game perfect equilibrium for this finite game is $\gamma_j^r = 0 \ \ \forall j$ and $M_i^s = 0 \ \ \forall i$. In this equilibrium, both types of players enforce their best response by sending/returning nothing, and this constitutes a non collective trust, and non collective trustworthiness equilibrium.

Now suppose that a negative economic shock destroys a fraction of receivers' endowment. Once again, assuming that agents follow an *homo economicus* rationale, each receiver maximizes his payoff by choosing γ_j^r and the reasoning would be the same: any positive value of γ_j^r diminishes his payoff, so he chooses to return no fraction of his endowment. Anticipating this strategic behavior, the sender optimizes his payoff described in Equation (2) by sending no positive amount $M_i^s = 0$. In the end, both types of players enforce their best response by sending nothing, and we end up in the same equilibrium as before: with no collective trust, and no collective trustworthiness. Under this assumptions, a negative economic shock should not change the behavior of any receiver or sender in the game.

2.2 *Homo Altruistic* Rationale

Although agents could be self-interested, there is strong evidence that people care about others (Arrow, 1972; Becker, 1974; Andreoni, 1990; Simon, 1992). Therefore, I consider another possible equilibrium between senders and receivers, that accounts for altruistic and prosocial behavior. In particular, let us consider a sender that cares about receivers' payoffs and derives disutility from unequal payoffs (Fehr and Schmidt, 1999; Bolton and Ockenfels, 2000). Under this scenario, the sender does not care only about his material payoff but instead tries to maximize the following inequality-averse utility function:[6]

$$\max_{M_i^s} U_i(M_i^s) = \pi_i^s(M_i^s) - \frac{\alpha}{2}\left(\pi_i^s(M_i^s) - \overline{\pi_j^r}(M_i^s)\right)^2, \tag{3}$$

where $\alpha \geq 0$ denotes the degree to which sender i is accountable for differences in final payoffs. Notice that if $\alpha = 0$, then the sender can be said as self-interested; on the other hand, if $\alpha > 0$ then the sender is said to be an inequality-averse agent. As seen in Equation (3), an inequality-averse sender derives less utility as long as his payoff differs from the average receiver's payoff. Moreover, disutility is not only a product of having more than others receivers (inequality aversion) but also feeling envy for having less than others.

[6] Although extensions for inequality-averse models may result in trivial modifications to the Fehr-Schmidt model (Engelmann, 2012), the inclusion of multiple agents from the same participant type on a trust game is not minor as problems for collective action may arise.

I assume that sender derives disutility by distancing himself from the *average* receivers' payoff for two reasons: First, because he cares about the entire group of receivers as a whole, and second, because he does not care more about one receiver than another in particular. Having said that, we define the average receiver's payoff as:[7]

$$\overline{\pi_j^r}(M_i^s) = \overline{E_j^r} + \left(\frac{\kappa}{R}\sum_{i=1}^{S} M_i^s\right)(1 - \overline{\gamma_j^r}). \tag{4}$$

Finally, let me assume that senders are the only inequality-averse agents in the model. Although receivers may show altruistic behavior, I focus on the effect of an economic shock that only affects the receiver's welfare, which makes more salient an advantageous inequality in favor of the senders.

2.3 Equilibria

In order to solve the optimization problem for an inequality-averse sender, let us first assume that he is an inequality-averse $\alpha > 0$ agent (which allows us to avoid corner solutions), and then consider the general case of $\gamma_j^r \in [0, 1]$.[8] With some algebraic arrangements, we can optimize Equation (3) which in equilibrium yields:[9]

$$M_i^{s^*} = \frac{\left(\frac{\kappa\overline{\gamma_r}}{S} - 1\right)}{\alpha\left(\frac{\kappa\overline{\gamma_r}}{S} - \frac{\kappa(1-\overline{\gamma_r})}{R} - 1\right)^2} - \frac{E_i^s - \overline{E_j^r}}{\left(\frac{\kappa\overline{\gamma_r}}{S} - \frac{\kappa(1-\overline{\gamma_r})}{R} - 1\right)} - \frac{\left(\sum_{m\neq i}^{S} M_m^s\right)\left(\frac{\kappa\overline{\gamma_r}}{S} - \frac{\kappa(1-\overline{\gamma_r})}{R}\right)}{\left(\frac{\kappa\overline{\gamma_r}}{S} - \frac{\kappa(1-\overline{\gamma_r})}{R} - 1\right)} \tag{5}$$

2.4 Predictions of the model

Equation (5) delineates the overarching behavior exhibited by an *homo altruistic* sender engaged in the collective trust game alongside other agents. Since the sender's optimal decisions hinge upon their beliefs regarding the behavior of others, it is imperative to delineate their optimal choices across various scenarios. For instance, let's examine a scenario where there is no economic inequality ($E_i^s = \overline{E^r}$), and the inequality-averse sender interacts (or perceives to interact) solely with agents embodying a *homo economicus* mindset. In such a setting, all individuals share identical economic circumstances and adhere to the rationality of *homo economicus*. Consequently, the inequality-averse sender anticipates zero contributions from their counterparts and expects trustworthiness levels to

[7] See Appendix A.3 for more details.

[8] It is possible to restrict $\gamma^r = 0$ assuming that the inequality averse sender interacts exclusively with *homo economicus* receivers but the equilibrium would not be characterized for the general scenario where positive values for trustworthiness and expectations of trustworthiness may appear.

[9] See Appendix B for more details.

remain at zero. Thus, in this scenario, we should observe that

Proposition 2.4.1. *Under equal endowments settings, and the presence of other homo economicus players, an inequality-averse and a non-inequality-averse sender should transfer the same amount $M_i^s = 0$ to the groups of receivers.*

Given the existence of an equal endowment setting $E_i^s = \overline{E_j^r}$ and the presence of *homo economicus* senders and receivers, the sender first internalize their behaviors and then optimize Equation (5). For his fellows senders, he assumes that $M_{i\neq j}^s = 0$, which in aggregate yields, $\sum_{m\neq i}^{S} M_m^s = 0$. With this restriction, the inequality-averse sender should send a positive amount ($M_i^s > 0$) if and only if the expectations of the average returned amount ($\overline{\gamma_j}$) by the group of receivers are greater than $\frac{S}{\kappa}$.

However, the latter scenario is not feasible, as receivers interact under an *homo economicus* rationale strategy. Therefore, receivers enforce their best strategy by playing $\gamma_j^r = 0$ for all j. Under these circumstances, there exists no positive amount sent that would optimize the inequality-averse sender's payoff. Consequently, the best response for the sender would be $M_i^s = 0$, resulting in the same *homo economicus* equilibrium. This outcome illustrates that there are no discernible differences in behavior between non-inequality-averse and inequality-averse senders in a scenario with equal endowments.

Let us now relax the assumption of interaction with *homo economicus* receivers, as there is evidence of positive trustworthiness in games of this nature (Johnson and Mislin, 2011). Additionally, let us formally restrict the average returned amount with the following condition:

Assumption 2.4.1: The average of the receiver's returned amount is lower than $\overline{\gamma_r} < \dfrac{S(R+\kappa)}{\kappa(S+R)}$,

which states that it is not feasible to systematically encounter a group of receivers with high levels of positive trustworthiness.[10]

Proposition 2.4.2: *If Assumption 2.4.1 holds, and there is an unequal endowment setting with presence of other homoeconomicus senders, then an inequality-averse sender should transfer more to the groups of receivers than a non-inequality-averse sender.*

Suppose that a negative economic shock destroys a fraction of the receivers' endowment, generating an inequality where $E_i^s > \overline{E^r}$. Notice that, once the shock occurs, the optimal solution presented in Equation (5) reflects the change in the transferred amount M_i^s when there is a reduction in the

[10] Note that this is not an assumption far from reality. For clarification, refer to a regular Trust Game (Berg et al., 1995) where $S = 1$, $R = 1$ and $\kappa = 3$. In that particular case, the average of the returned amount by receivers must fulfill to be less than $\frac{2}{3}$. If we review the data presented by Johnson and Mislin (2011), we can observe that not only the returned amount is less than $\frac{2}{3}$, but the expectations are also less than $\frac{2}{3}$. In particular, for an scenario of $S = 2$, $R = 2$, and $\kappa = 4$, then $\overline{\gamma_r}$ must be less $\frac{3}{4}$.

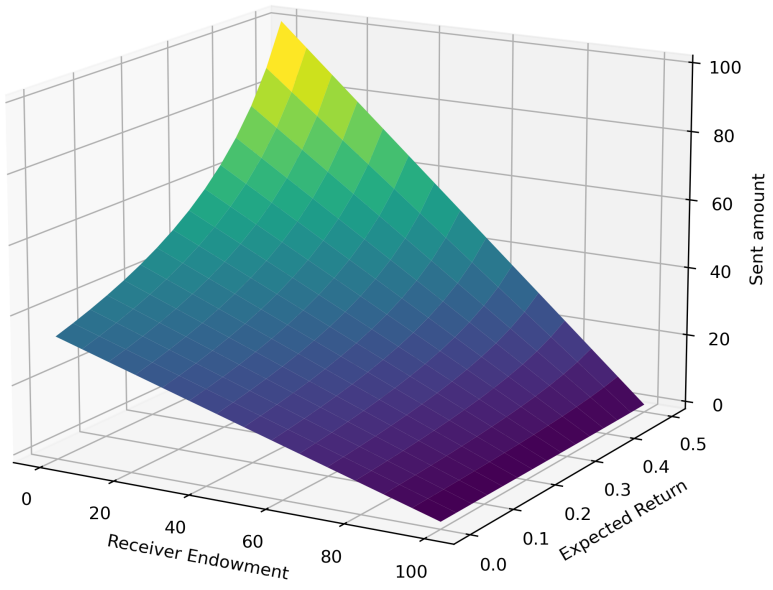
average receiver's payoff, and we can characterize a formal expression for these changes as,

$$\frac{\partial M_i^s}{\partial \overline{E_j^r}} = \frac{1}{\left(\frac{\kappa \overline{\gamma_r}}{S} - \frac{\kappa(1-\overline{\gamma_r})}{R} - 1\right)} \tag{6}$$

Given that Assumption 2.4.1 holds, Equation (6) is negative, which represents that an economic shock that reduces the average of the receivers endowment should increase the transferred amount. Furthermore, this has behavioral incentives in addition to he theoretical ones, since altruistic agents might be motivated to repair receivers following a shock in order to experience a "'warm-glow" effect (Andreoni, 1990), so expectations of the returned amount may be lower too. Figure 2 displays a hypothetical altruistic behavior.

From Figure 2, notice that an allocation of equal endowments leads to a transfer of zero funds, and this optimal choice is independent of the beliefs about the trustworthiness of the group of receivers. Then, as the inequality generated by the shock increases, so does the amount sent by the sender. Finally, note that the increase is much more accelerated as beliefs about the average returned amount increase. This means that, once we relax the assumption of interaction with *homo economicus* receivers, the inequality-averse sender has incentives to increase the sent amount, as the expected return amount may increase.

Figure 2: Inequality-averse sender Behavior



Note: the *x-axis* denotes the average of the receivers endowment ($\overline{E^r}$), the *y-axis* denotes the expected average of returned amount ($\overline{\gamma^r}$), and the *z-axis* denotes the sent amount by the altruistic sender (M_i^s). In violet = low levels of sent amount; in yellow = high levels of sent amount. The following parameters were fixed at: $R = 2$, $S = 2$, $\alpha = 1$, $\kappa = 4$, $M^s_{m \neq i} = 0$.

Furthermore, it is also possible to relax the assumption of interaction with *homo economicus* senders. For instance, let us consider the scenario where expectations about other senders contributions are not zero $\left(\sum_{m\neq i}^{S} M_m^s \neq 0\right)$.

Proposition 2.4.3. *If Assumption 2.4.1 holds, for any particular setting of endowments $(E^s, \overline{E^r})$, there is a $\overline{\gamma^*}$ that motivates an inequality-averse sender to transfer* ***less*** *as his expectations about other senders contributions decreases.*

It is possible to prove this proposition by estimating the marginal change in the transferred amount M_i^s when there are changes in the expectations about other senders contributions $\left(\sum_{m\neq i}^{S} M_m^s\right)$. To do so, we can characterize a formal expression for these changes as,

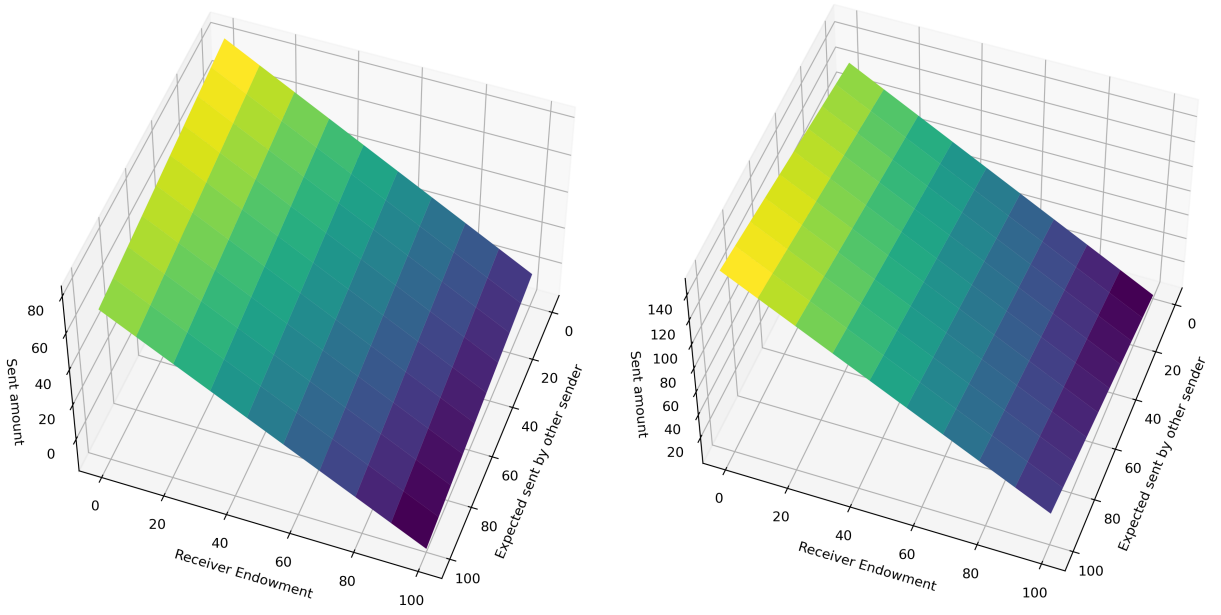
$$\frac{\partial M_i^s}{\partial\left(\sum_{m\neq i}^{S} M_m^s\right)} = -\frac{\left(\frac{\kappa\overline{\gamma_r}}{S} - \frac{\kappa(1-\overline{\gamma_r})}{R}\right)}{\left(\frac{\kappa\overline{\gamma_r}}{S} - \frac{\kappa(1-\overline{\gamma_r})}{R} - 1\right)}. \tag{7}$$

Given that Assumption 2.4.1 holds, then the direction of the derivative presented in Equation (7) is determined by the numerator, and in particular by the value of the expected returned amount ($\overline{\gamma_r}$). As can be seen, there is a critical value at $\overline{\gamma^c} = \frac{S}{(S+R)}$, where the optimal decision of the sender does not change when his expectations about other sender's contributions changes, and this critical value is consistent with an equitable distribution of resources. Then, as the expected returned amount increases $\overline{\gamma_r} > \overline{\gamma^c}$, the inequality-averse sender begins to experience positive returns from the contributions of others, which makes it more profitable to trust the group of receivers and leads him to increase his contribution. On the other hand, when the expected returned amount is lower than the critical value $\overline{\gamma_r} < \overline{\gamma^c}$, there are incentives to free-ride on the contributions of others, which leads to a reduction in the transferred amount. Finally, notice that this relationship is independent of the distribution of endowments so we should observe this behavior independently on the occurrence of an economic shock. Formally, including Assumption 2.4.1, I can express this $\overline{\gamma^*}$ as $\left(\frac{S(R+\kappa)}{\kappa(S+R)} > \overline{\gamma^*} > \frac{S}{(S+R)}\right)$.

In Figure 3a, I show the latter relationship when $\overline{\gamma_r} < \overline{\gamma^c}$, which means that the expected returned amount is lower than the cutoff. Note that as the expectations about the contribution of other senders increase, the optimal solution is to reduce the sent amount and free-ride on the contributions of others.[11] Conversely, Figure 3b shows the latter relationship when $\overline{\gamma_r} > \overline{\gamma^c}$ and, as can be seen, higher contributions from the inequality-averse sender occur when his expectations about other senders contribution increases. Therefore, lower expectations in the contribution of other senders should reduce the amount sent by the inequality-averse sender.

[11] Darker colors represent lower levels of the amount sent (denoted as M_i^s). As can be seen, lower sent amounts occur when the sender beliefs that others would transfer their entire endowment.

Figure 3: Altruistic behavior across beliefs $\left(\overline{\gamma^r}, \sum_{m\neq i}^{S} M_m^s\right)$



(a) (**Low**) Expected average returned amount $\overline{\gamma_j} \in [0, \overline{\gamma^c})$

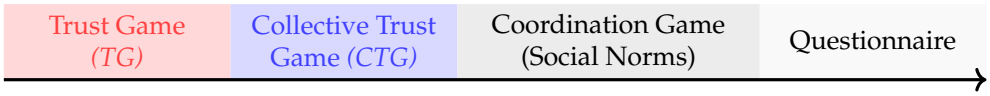
(b) (**High**) Expected average returned amount $\overline{\gamma_j} \in \left[\overline{\gamma^c}, \frac{S(R+\kappa)}{\kappa(S+R)}\right]$

Note: the *x-axis* denotes the average of the receivers endowment ($\overline{E^r}$), the *y-axis* denotes the expected sent amount by other senders ($\sum_{m\neq i}^{S} M_m^s$), and the *z-axis* denotes the amount sent by the inequality-averse sender (M_i^s). **Left Figure 3a:** has an expected returned amount of $\overline{\gamma^r} = 0.45$. **Right Figure 3b:** has an expected returned amount of $\overline{\gamma^r} = 0.55$. In violet = low levels of sent amount; in yellow = high levels of sent amount. The following parameters were fixed at: $R = 2$, $S = 2$, $\alpha = 1$, $\kappa = 4$. Finally, $\overline{\gamma^c} = 0.5$.

3 Experimental Design

Given these theoretical results, and in order to provide empirical evidence, I implement a laboratory experiment. The experiment was conducted using oTree (Chen et al., 2016) involving 476 college students from Bogotá, Colombia. Participants were recruited through the Rosario Experimental and Behavioral Economics Lab (REβEL, Greiner (2015)) at the Universidad del Rosario between November 11, 2023 to February 26, 2024. During the experiment, subjects participate in three different activities. The timeline of the experiment is depicted in Figure 4. It is worth mentioning that all treatments were ran with the same timeline, subjects were informed that only one random activity would be paid out, and the average duration per session was one hour and twenty minutes.

Figure 4: Experimental design timeline



3.1 Trust Game (TG)

In the first part of the experiment, participants engage in a one-shot trust game (Berg et al., 1995) with no feedback and with a multiplier of $\kappa = 4$.[12] During the interaction, all subjects started with an endowment of 100 experimental points (EP),[13] and this information was common knowledge. Subsequently, each participant was randomly assigned to one role: **sender** or **receiver**, and this role was fixed during the entire session. As soon as they received their role, each participant was paired with a subject of a different type. Then, the sender decides how much of his endowment he wants to transfer to the receiver, ranging from 0, 20, 40, 60, 80, to 100% of his endowment. On the other hand, the receiver decides how much of the transferred amount between 0 and the total transfer he wants to return to the sender, with this choice being continuous. For the receiver's choice, I use the strategy method which allows me to recover the receiver's returned amount function. In addition, each participant responded to an incentive-compatible expectation task about the other's player behavior, including the expected returned amount by the receiver and the expected sent amount by the sender. Each player received 5EP each time their expectation matched their partner's response.

3.2 Collective Trust Game (CTG)

The main activity of the experiment consists of a modified version of the collective trust game (CTG) presented by Holm and Nystedt (2010), with the particularity that now each sender decides privately how much of his endowment he wants to send to the group of receivers, following the structure presented in Section 2.[14] During the interaction, all subjects started with an endowment of 100 EP, and given that all players start with an equal endowment and it is common knowledge, the sender's primary motivation for sending money to the group of receivers is his belief that receivers will return the favor in the second stage of the game. Likewise, any positive amount returned by the group of receivers to the group of senders is taken as collective trustworthiness.

In this setup, participants are randomly assigned into groups of **four** people (two per role). Once grouped, each sender decides how much of his endowment he wants to transfer to the **group** of receivers, ranging from 0, 20, 40, 60, 80, to 100% of his endowment. Then, the total of transfers are summed, multiplied by 4, and divided equally among the group of receivers. On the other hand, each receiver decides simultaneously how much of the transferred funds he wants to return to the

[12] It is not common to play a Trust Game with a multiplier different from $\kappa = 3$ (Johnson and Mislin, 2011). Given that I employed a multiplier of $\kappa = 4$ at the Collective Trust Game, I also implemented a two-person Trust Game with the same multiplier to compare outcomes fairly.

[13] This is equivalent to 20000 COP (5 USD). Each experimental point (EP) was valued at a fee of $200 pesos ($0.05 USD). Participants received $12,000 pesos ($3 USD) as a show-up fee. On average, participants received a payment of $40,000 pesos ($10.8 USD). Given that the minimum wage in Colombia is $5,531 pesos/hour ($1.42 USD/hour), the earnings obtained are about 7.2 times the minimum wage.

[14] In Holm and Nystedt (2010), senders (trustors) choose whether to (I) "invest" or (N) "not to invest" on receivers (trustees) by voting (I) or (N). If the majority of trustors decided to invest, then the trustees had to vote between (R) "reciprocating" or (D) "defecting" and, again, the majority rule applies to decide the trustees' decision.

group of trustors, ranging between 0 and the total transferred funds received.[15] For the receiver's choice, I use the strategy method to recover the receiver's trustworthiness function. Once receivers made their choices, all fractions of the transferred amounts are totaled up and divided equally among the group of senders. The structure of the game is presented in Table C1 and Figure C1 in the Appendix.

3.3 Treatment arms

Participants repeated the collective trust interaction twice and were re-matched in groups with random subjects for both rounds. Moreover, the outcomes of each round were not revealed until the end of the experiment. Given that they interact twice, we define the following two treatments:

***Control* group**: In this group, all participants, regardless of their role, played the Collective Trust Game maintaining an endowment of 100 EP during both rounds. Furthermore, there is no mention of negative shocks or the possibility of a change in the initial setting.

***Treatment* group**: In this group, all participants played the Collective Trust Game with an endowment of 100 EP during the first round, and then the **negative economic shock** (NES) takes place. Specifically, at the beginning of the second round, all subjects were informed about a negative income shock that destroyed each **receiver's endowment** by 60%, reducing it from 100 EP to 40 EP.

Notice that by not giving feedback in the previous round, I ensure that the measure of the outcomes in the second round is the causal effect of the negative economic shock, and not result from specific previous outcomes. Table 1 depicts an overview of the distribution of participants across treatments and roles.

Table 1: Distribution of participants by role and treatment

	Senders	**Receivers**	**Total**
Control	120	120	240
Treatment	118	118	236
Total	**238**	**238**	**476**

In addition, participants were asked their first-order beliefs about others' behavior. For instance, senders were asked to predict the average sent amount by their partner and the average returned amount by the group of receivers (as a percentage). In the same way, receivers were asked to predict the senders' average sent amount and their partner's average returned amount (as a percentage).

[15] The receiver's transfer is the total transfer divided by 2.

Both measures were incentive-compatible expectations given that each participant earned 5 EP each time they predicted their partner's decision.

3.4 Coordination Game (Social Norms)

Finally, the experimental subjects engage in a Krupka and Weber (2013) coordination game, which is employed to elicit incentive-normative expectations from senders and receivers during the first round of the CTG.[16] Each participant was required to rate the degree of social appropriateness of the sender's and receiver's behavior. Hence, each subject must indicate whether sending or returning a potential amount is extremely improper socially, somewhat unsuitable socially, somewhat socially appropriate, or very socially appropriate. Each subject earn 5 EP each time their normative expectation match the expectation of another subject of his own same type. At the end of the activity, the participants completed a final socio-demographic questionnaire and requested to fill in information for payment.

4 Results

4.1 Sample characteristics

Table 3 presents the balance between the treatments arms. Pooling participants in the first round of the Collective Trust Game, senders transfer on average 43% (SD = 0.28) of their initial endowment, with a median of 40%. Moreover, the fraction of senders that send nothing (10.9%) is equal to the fraction of senders that send their full endowment (10.4%). The distribution of the sent amount is presented in the left panel of Figure D1 in the Appendix.

Regarding collective trustworthiness, each receiver decides how much he wants to return to the group of receivers as a fraction of the total transfer. Given that I employed the strategy method, I calculate the *average returned amount* as the average of fractions returned.[17] On average, in the round before the shock, the returned amount is 29% (SD = 0.18) of the transferred funds. Furthermore, 7% of the receivers return nothing, nearly 5% return 25% of the total transfer and 12% of them return the 50%. Left panel of Figure D2 in the Appendix depicts the distribution of the average returned amount in the first round of the interaction.

[16] The coordination game was implemented at the end of the experiment, and the identity of the players remained anonymous. We recall all treatments to situate themselves in the first round, where all participants had an equal endowment of 100 EP.

[17] Given that each sender has the possibility to transfer 0, 20, 40, 60, 80 or 100 points, then the **group** of senders could transfer 10%, 20%, 30%, ... , 100% of the total endowment. Notice that if only one sender sends 20 points, and both senders have 100 points (200 points in total), then the group transfer is 10% of the total endowment. The strategy method recover all 10 possible transfers.

Table 3: Balance between treatments: Control and Receiver's Shock

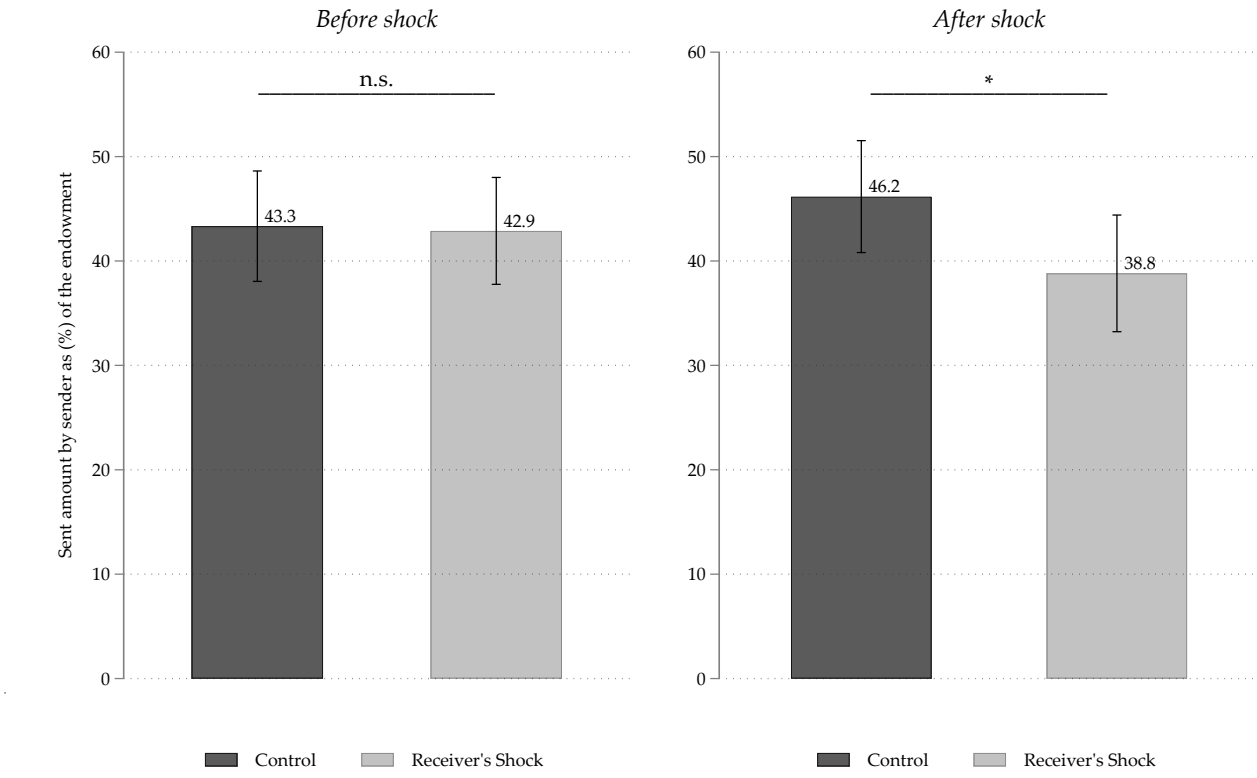
Variable	Mean **Control** (240 obs)	Mean **Treatment** (236 obs)	*p-value* for H_0 Control = Treatment
Panel A: Socio-demographic variables			
Age	22.09	22.03	0.85
Gender	0.60	0.58	0.68
Siblings	1.56	1.51	0.67
Stratum	3.14	3.20	0.54
Self-perception of wealth (income stair)	5.59	5.61	0.86
Income (USD)	84.00	81.24	0.82
Expenses (USD)	44.01	53.98	0.16
Semester	5.25	5.32	0.82
Master's student	0.05	0.07	0.54
Panel B: Attitudes and Preferences			
Sent amount (percentage) in TG (Activity 1)	39.17	43.05	0.30
Generalized Trust Measure (0: Less - 10: More)	4.30	4.41	0.60
Generalized Risk Measure (0: Less - 10: More)	6.42	6.21	0.24
Rationality Measure	36.13	35.60	0.80
Have you participated in experiments before?	0.75	0.76	0.83
Importance of religion (1: Less - 4: More)	2.42	2.33	0.28
Medical care (0: State - 10: Patient)	3.94	3.52	0.06*
Political preference (1: Right - 5: Left)	3.68	3.71	0.80
Welfare opinion (1: Less - 5: More)	3.22	3.22	1.00
Inequality opinion (1: Less - 5: More)	4.03	4.08	0.61

Note: **Self-perception of wealth**: is a categorical variable that denotes the participant's answer to the question: "Imagine a staircase with 10 steps. People with the lowest incomes are placed on step 1 and people with the highest incomes on step 10. Where would you place your family group?". **Rationality measure**: denotes the participant's answer to the beauty contest of *Guess 2/3 of the average*. **Medical care**: is a variable that indicates between 0 (State) and 10 (Patient) who should paid more for a medical treatment. **Welfare opinion**: is a categorical variable that ranges from 1 = "Totally disagree" to 5 = "Totally agree" to the statement: "Each individual is responsible for their own well-being". **Inequality opinion** is a categorical variable that ranges from 1 = "Totally disagree" to 5 = "Totally agree" to the statement: "The government should implement strong policies to reduce inequality". $^{***}p < 0.01$,$^{**}p < 0.05$,$^{*}p < 0.1$

4.2 Collective Trust Behavior

Initially, I analyze the effect of the negative economic shock over collective trust, quantified as the percentage of the amount sent relatively to the initial endowment. Figure 5 depicts the average transfer amount from senders to the group of receivers by treatment. In the left panel, the average sent amount in the round before the shock for the control group is 43.3%, and for the treatment group is 42.9%, with no statistically significant difference between both treatments (p-value = 0.903). This suggest that there is "pre-treatment" balance between groups. In the right panel, we observe the effect of the negative shock over the treatment group. Specifically, we observe an 8*pp* reduction (equivalent to a 16% decrease) compared to the control group's average of 46%. The difference is statistically significant at 10% level. (p-value = 0.062), suggesting a discernible effect of the shock on trust.

Figure 5: Sent amount in the Collective Trust Game



Note: Receiver's shock (**Treatment**). **Treatment**: 60% reduction in the receiver's initial endowment. Sender's initial endowment remain = 100 EP for both treatments and rounds. 95% clustered confidence intervals at participant level. **Left panel:** Denotes the sent amount in the round before the shock. **Right panel:** Denotes the sent amount in the round after the shock. No controls included. $^{***}p < 0.01$,** $p < 0.05$,* $p < 0.1$, n.s $p > 0.1$

4.2.1 Regression Analysis

In order to measure causal treatment effects on collective trust behavior, I estimate the following model:

$$Sent_i = \alpha_0 + \alpha_1 Shock_i + x'_{1i}\beta + x'_{2i}\gamma + \varepsilon_i, \tag{8}$$

where $Sent_i$ denotes the proportion of the endowment that sender i sends to the group of receivers. $Shock_i$ is a dummy variable that takes value of 1 if the participant was treated and 0 otherwise. Additionally, I include a set of controls (x_{1i}) that consider sociodemographic variables, and I also include the trustworthiness belief's (x_{2i}) elicited within the round, given that trust levels may be affected by the expected returned amount (Ashraf et al., 2006). Considering that there is no feedback of experimental outcomes during the interaction, there is no reason to think that auto correlation within the group occurs. Therefore, the standard errors (ε_i) are clustered at individual level (robust standard errors) (Abadie et al., 2023).

Table 4 depicts the results of the OLS estimation for both rounds. The dependent variable for all specifications is the sent amount by the sender. In columns 1 to 3, I estimate the treatment differences in the round before the shock, where we expect no statistical differences between treatments. Columns 4 to 6 are the estimates in the round after the shock. Columns 1 and 4 depict unconditional means before and after the shock (which are the estimates from Figure 5).

Table 4: Collective **Trust** - Main Results

Dep Var: **Sent amount**	Round					
	Before shock			After shock		
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment	-0.45	-2.06	-2.02	-7.35*	-8.91***	-5.80*
	(3.72)	(2.99)	(2.93)	(3.91)	(3.25)	(3.19)
Constant	43.33***	13.74	-1.88	46.17***	20.83	-0.33
	(2.67)	(14.24)	(13.60)	(2.71)	(14.53)	(14.49)
Mean Dep. Variable	43.3	43.3	43.3	46.2	46.2	46.2
Effect size (%)	-1.0	-4.8	-4.7	-15.9	-19.3	-12.6
Controls		✓	✓		✓	✓
Beliefs controls			✓			✓
Observations	238	238	238	238	238	238
Adjusted R-squared	0.00	0.38	0.43	0.01	0.35	0.43

Note: The set of controls included in ***Controls*** are: age, gender, stratum, risk measure, the sent amount in regular trust game, previous experience participating in experiments, and medical care -as it shows unbalance at 10% level (p-value = 0.06)-. The set of controls included in ***Beliefs*** controls are the expected returned amount by the sender for each possible transferred amount (strategy method). The *Mean Dep. Variable* express the unconditional control mean of the dependent variable. The *Effect size (%)* denotes the size of the treatment effect as a percentage compared with the unconditional mean of the control group. Standard errors clustered at individual level in parenthesis. $^{***}p < 0.01, ^{**} p < 0.05, ^{*} p < 0.1$

From columns 1 to 3, we observe that there are no differences in the average sent amount, and this balance between treatment groups is robust to inclusion of a set of controls. I add several controls including participant's age, whether the participant is female, the socioeconomic stratum, the participant's preference for risk measure, whether the participant has participated in experiments before, the medical opinion (given the unbalance presented between treatments), and the participant's sent amount in the trust game. Moreover, I also add as controls the participant's beliefs in the returned amount by the group of receivers for each possible transfer (strategy method)[18] and the point estimates show no statistical variation.

Regarding the effect of the shock over trust, observe columns 4 to 6. In column 4, the point estimate of -7.3*pp* shows a reduction in the average sent amount of 15% compared to the control group.

[18] This result is robust to include *beliefs* control as the **average** of returned amount across possible transfers instead of each possible returned amount for each transfer.

When the first set of controls is included, the point estimate decreases to -8.9*pp*, making more salient the effect of the economic shock. Finally, when I include the sender's beliefs about the returned amount, the treatment effect reflects a -5.8*pp* decrease in trust, and it remains statistically significant at 10% level (p-value = 0.065). This reduction in collective trust can be explained by an increase in the percentage of senders that are transferring 0% of their endowment (17% of senders in the treatment group while 11% of senders in the control group) or 20% of their endowment (31% of senders in the treatment group while 18% of senders in the control group). The right panel of Figure D1 in the Appendix shows the distribution of the sent amount in the round after the shock.

4.3 Collective Trustworthiness behavior

Regarding collective trustworthiness behavior, I analyze the effect of the negative economic shock over collective trustworthiness, quantified as the percentage of the returned amount by each receiver relatively to the receiver's transfer.[19] Figure 6 depicts the average returned amount from receivers to the group of senders across all possible transfers. In particular, the black bar (10% transfer) shows the average returned amount when the group of senders sent 20 points (10% of the total endowment of 200 points per group), the group of receivers receive 80 points (20 points $\times$ 4), and each receiver obtains 40 points. Conversely, the latter grey bar (100% transfer) shows the same variable when the group of senders sent the entire endowment of 200 points, and each receiver obtains 400 points.

The left panel of Figure 6 shows that the average returned amount in the round before the shock is close to 30% of the transfer, independently on the transferred amount by senders. The absence of a positive slope between the quantity returned as a percentage and the total transferred amount shows that there is no positive reciprocity under an equal endowment setting. Moreover, both of the treatments depict this same behavior in the round before shock, which shows it is a general pattern (Figure D3 in the Appendix). However, in the right panel we observe that the average returned amount remains at 30% for the control group, and the absence of positive reciprocity remains, while for the treatment group there is a general reduction close to -10*pp* for all possible transfers, which suggest a negative effect of the shock on collective trustworthiness.

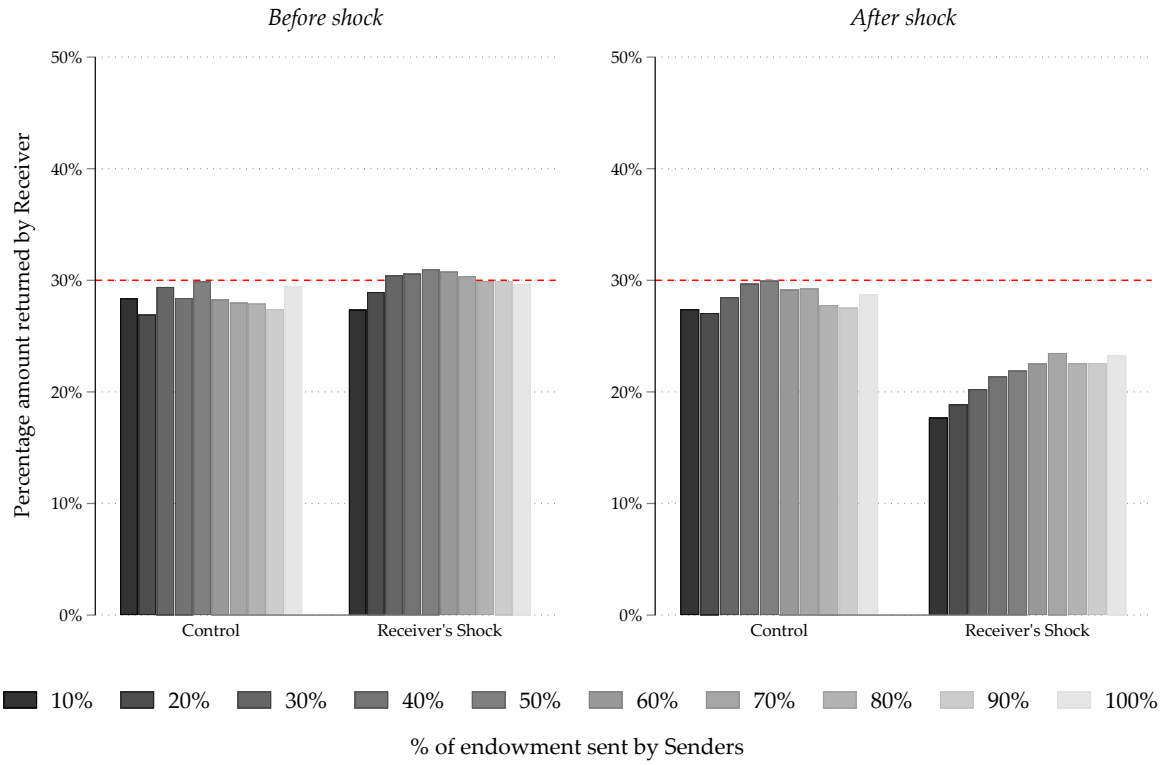
4.3.1 Regression Analysis

In order to measure causal treatment effects on collective trustworthiness behavior, I estimate the following model:

$$Returned_j^m = \alpha_0 + \alpha_1 Shock_j + x_{1j}'\beta + x_{2j}'\phi + \varepsilon_j, \tag{9}$$

[19] The receiver's transfer is the total transfer divided by 2.

Figure 6: Returned amount in the Collective Trust Game



Note: Receiver's shock (**Treatment**). **Treatment**: 60% reduction in the receiver's initial endowment. Sender's initial endowment remain = 100 EP for both treatments and rounds. Receiver's initial endowment = 100 EP for both treatments. Receiver's treatment endowment in the second round = 60 EP. **Left panel:** Denotes the percentage amount returned in the round before the shock. **Right panel:** Denotes the percentage amount returned in the round after the shock. Given the strategy method, receiver respond: How much do you want to return if senders sent ... % of their endowment. Each bar denotes one of the 10 responses. No controls included.

where $Returned_j^m$ denotes the proportion of the transfer that receiver j returns to the group of senders for a particular sent amount m. The dummy variable $Shock_j$ denotes the treatment assignation. I employ the same set of controls (x_{1j}) including participant's age, whether the participant is female, the socioeconomic stratum, the participant's preference for risk measure, whether the participant has participated in experiments before, the medical opinion (given the unbalance presented between treatments), and the participant's returned amount in the trust game. For the *beliefs* controls (x_{2j}) I include the trust empirical expectation elicited within the round, as trustworthiness may be affected by the expected sent amount. Finally, standard errors are clustered at individual level.

Table 5: Collective **Trustworthiness** - Main Results

Dep Var: **Returned amount** *(as percentage of the transferred)*	*Strategy Method*: Percentage of transfer by the group of senders									
	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
Panel A: Round *before* shock										
Treatment	-0.01	0.03	0.01	0.03	0.02	0.03	0.03	0.02	0.03	0.01
	(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Constant	-0.07	-0.10	-0.07	-0.04	-0.13	-0.10	-0.08	-0.11	-0.10	-0.10
	(0.11)	(0.09)	(0.09)	(0.10)	(0.09)	(0.10)	(0.10)	(0.10)	(0.10)	(0.10)
Mean Dep. Variable	0.28	0.27	0.29	0.28	0.30	0.28	0.28	0.28	0.27	0.29
Effect size (%)	-2.1	9.3	4.7	9.1	5.5	9.7	9.9	8.8	11.2	2.0
Controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Beliefs controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	238	238	238	238	238	238	238	238	238	238
Adjusted R-squared	0.18	0.20	0.19	0.16	0.19	0.15	0.14	0.16	0.16	0.17
Panel B: Round *after* shock	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
Treatment	-0.08***	-0.07***	-0.07***	-0.07***	-0.07***	-0.05**	-0.05**	-0.04*	-0.04*	-0.04*
	(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Constant	0.01	-0.09	-0.08	-0.12	-0.08	-0.03	-0.06	-0.10	-0.13	-0.14
	(0.10)	(0.09)	(0.09)	(0.09)	(0.09)	(0.11)	(0.09)	(0.10)	(0.09)	(0.10)
Mean Dep. Variable	0.27	0.27	0.28	0.30	0.30	0.29	0.29	0.28	0.28	0.29
Effect size (%)	-28.5	-24.9	-23.5	-23.9	-23.6	-18.7	-15.5	-14.2	-13.6	-13.5
Controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Beliefs controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	238	238	238	238	238	238	238	238	238	238
Adjusted R-squared	0.19	0.21	0.22	0.25	0.21	0.19	0.19	0.18	0.19	0.22

Note: The set of controls included in ***Controls*** are: age, gender, stratum, risk measure, the returned amount in the regular trust game for each possible transfer (strategy method), previous experience participating in experiments, and medical care -as it shows unbalance at 10% level (p-value = 0.06)-. The set of controls included in ***Beliefs*** controls are the receiver's expected sent amount by the group of senders. The *Mean Dep. Variable* express the unconditional control mean of the dependent variable. The *Effect size (%)* denotes the size of the treatment effect as a percentage compared with the unconditional mean of the control group. Standard errors clustered at individual level in parenthesis. $^{***}p < 0.01, ^{**}p < 0.05, ^{*}p < 0.1$

Table 5 depicts the OLS estimates for the Equation (9). The dependent variable for column 1 shows the percentage of the returned amount when senders sent 10% of their endowment. Likewise, column 10 shows the percentage of the returned amount when senders sent 100% of their endowment. Additionally, Panel A shows the treatment differences in the round before the shock, while Panel B shows the treatment differences in the round after the shock. I expect no differences in the average returned amount between groups in Panel A.

From Panel A, we observe that there are no differences between treatments in the average returned amount for each possible transfer. Notice that receivers in the control group returns in average 30% of the received transfer, and there are no differences in the returned amount across possible transfers by senders. Moreover, this balance between treatment groups is robust to inclusion of a set of controls that includes participant's age, gender, the socioeconomic stratum, the participant's preference for risk measure, whether the participant has played in experiments before, the medical opinion (given the unbalance presented between treatments), and the participant's returned amount in the trust game. Finally, I also add as *beliefs* controls the receiver's belief in the sent amount by the group of senders and the point estimates show no statistical variation.

Regarding the effect of the shock over trustworthiness, observe Panel B. In column 1, the point estimate of -0.08 reflects a reduction of 8*pp* in the collective trustworthiness, when the group of senders sent 10% of the total endowment. From columns 2 to 5, we observe that the economic shock reduces between 30% and 20% the average returned amount, and all the effects are statistically significant at 1% level. These results shows that, when the group of senders sent less than 50% of their endowment, the reduction in trustworthiness is more severe. Finally, columns 5 to 10 depicts reductions in the average returned amount ranging between 18% and 13%, which shows that even if senders sent their entire endowment (100%), the returned amount is lower once the shock occurs.

Similarly, this general reduction is motivated by an increase in the percentage of receivers that are returning between 0% and 25% of their transfer. In particular, the average of the returned amount across all possible transfers in the round after shock for the control group is 28% (SD = 0.18, max = 0.70) while for the treatment groups is 21% (SD = 0.15, max = 0.58). The right panel of Figure D2 in the Appendix shows the kernel density of the average of the returned amount across all transfers in the round after the shock disaggregated by treatments.

4.3.2 Positive reciprocity in the returned amount

If receivers return larger amounts when the transferred amount increases, then this is taken as evidence of positive reciprocity (Aycinena and Blanco, 2022). Although left panel of Figure 6 depicts little evidence of this behavior in the round before the shock, the right panel shows an increase in the returned amount across the amount sent by senders once the shock occurs. To corroborate the

presence of positive reciprocity in the receiver's behavior, I estimate the following model,

$$AverageReturn_j = \alpha_0 + \alpha_1 Shock_j + \alpha_2 SentAmount + \tau(Shock_j \times SentAmount) + x'_{1j}\beta + \varepsilon_j, \quad (10)$$

where $AverageReturn_j$ denotes the average returned amount by receiver j across all possible transfers, $Shock_j$ denotes the treatment assignation, $SentAmount$ is a continuous variable that denotes the sender's transfer allocation, ranging from 20%, 30%, to 100%, and (x_{1j}) denotes the set of controls for trustworthiness behavior. The parameter of interest is recovered in τ for the heterogeneous treatment effect across sent amounts. Moreover, the presence of positive reciprocity should exhibit a positive coefficient for larger transferred amounts.

The OLS estimates for Equation (10) are depicted in Table 6. Notice from Table 6 that the negative economic shock still significantly impacts collective trustworthiness. However, once the shock occurs, larger transferred amounts are followed by an increase in the returned amount. Figure 7 depicts the general behavior for collective trustworthiness.[20]

Table 6: Positive Reciprocity in **Trustworthiness** - Main Results

Dep Var: **Average amount returned**	Round			
	Before shock		After shock	
	(1)	(2)	(3)	(4)
Treatment	0.00771	0.00705	-0.09890***	-0.10161***
	(0.027)	(0.025)	(0.026)	(0.024)
Amount sent	0.00002	0.00002	0.00006	0.00006
	(0.000)	(0.000)	(0.000)	(0.000)
Treatment × Amount sent	0.00013	0.00013	0.00052**	0.00052**
	(0.000)	(0.000)	(0.000)	(0.000)
Constant	0.28310***	0.01020	0.28175***	0.01253
	(0.019)	(0.099)	(0.020)	(0.096)
Mean Dep. Variable	0.3	0.3	0.3	0.3
Effect size (%)	2.7	2.5	-34.7	-35.7
Sociodemographic controls		✓		✓
Observations	2380	2380	2380	2380
Clusters	238	238	238	238
Adjusted R-squared	0.00	0.07	0.04	0.11

Note: The set of controls included in ***Controls*** are: age, gender, stratum, risk measure, previous experience participating in experiments, and medical care -as it shows unbalance at 10% level (p-value = 0.06)-. Clustered standard errors by participant in parenthesis. $^{***}p < 0.01$,$^{**}p < 0.05$,$^{*}p < 0.1$.

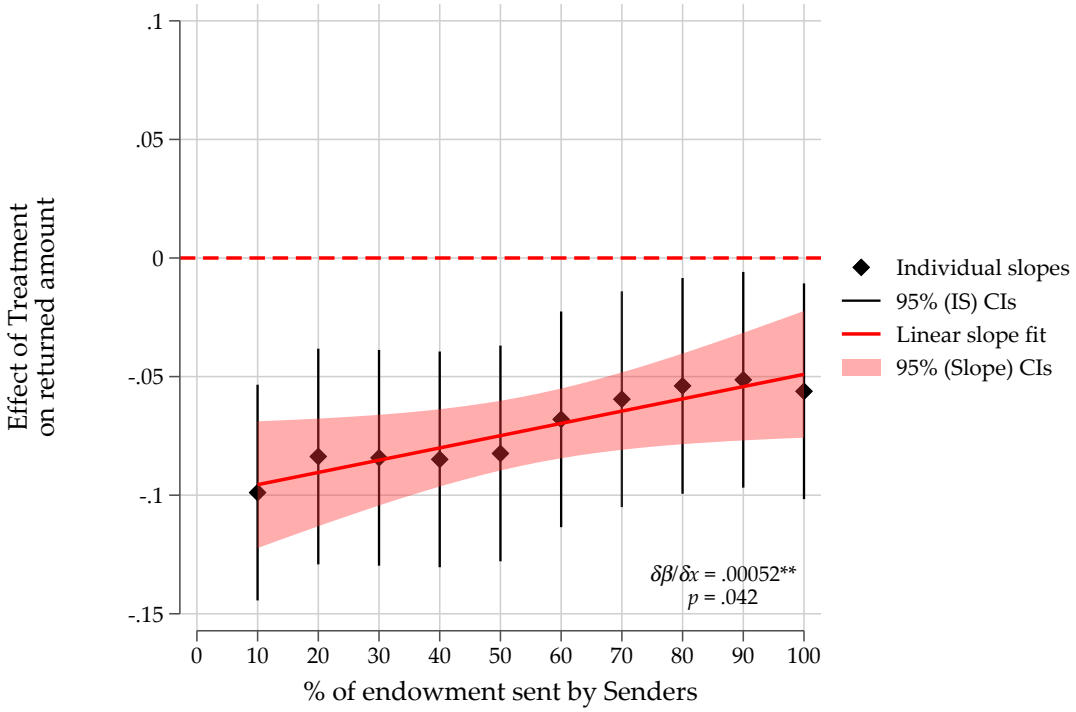
[20] While the conditional marginal treatment effect assumes a constant relationship across transferred amounts, I also estimate marginal heterogeneous treatment effects for each possible transfer using the following model,

$$AverageReturn_j = \alpha_0 + \alpha_1 Shock_j + \sum_m \alpha_m SentAmount_m + \sum_m \tau_m Shock_j \times SentAmount_m + x'_{1j}\beta + \varepsilon_j, \quad (11)$$

The OLS estimates for this specification form are presented in Table D1 in the Appendix.

The individual slopes (diamonds) denote the treatment effect for each possible transfer, which is the direct estimation of Equation (9). Observe that the point estimates from the individual slopes are not statistically different from the ones presented in Table 5, but are numerically different as they do not include the *beliefs* control, but overall denotes the harmful effect of the shock. Conversely, the linear slope (solid line) plots the conditional marginal treatment effect across all transfers. The linear slope (τ) shows a positive and statistically significant increase in the returned amount when the transferred amount by senders increase (p-value = 0.042). In general, this behavior show that receivers tend to return less once the shock takes place, being this negative effect more pronounced when senders transfer low allocations. However, receivers exhibit positive reciprocity, as an increase of 40 points in the transferred amount translates in an increase of 2*pp* in the returned amount. Finally, Figure D3 in the Appendix depicts the estimation for positive reciprocity in the round before the shock, showing the absence of this phenomena before the shock.

Figure 7: Positive reciprocity in the returned amount - *After shock*

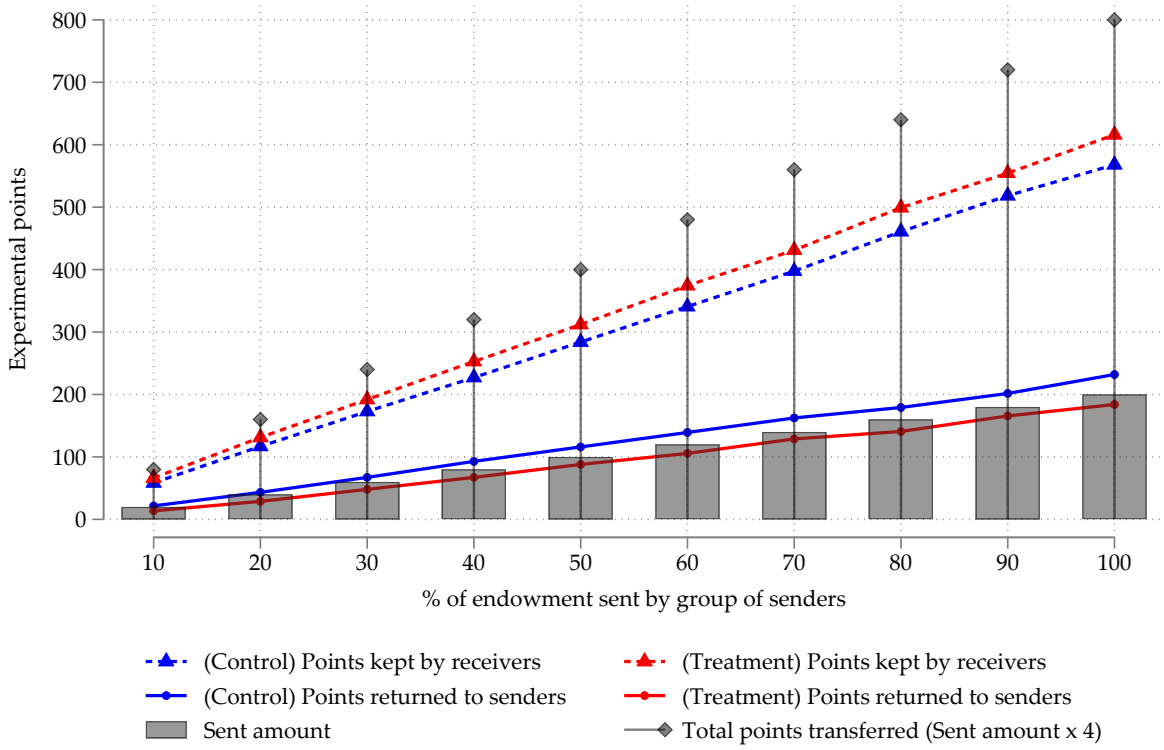


Note: Conditional marginal treatment effect over trustworthiness. Linear trend presented in red. Confidence interval for the linear slope at 95% level. In diamonds, individual treatment effects for each possible transferred amount. 95% confidence intervals for individual slopes (treatment effects) are depicted in black. Finally, $\frac{\partial \beta}{\partial x}$ denote the constant marginal change across transferred amounts. The set of controls included are: age, gender, stratum, risk measure, previous experience participating in experiments, and medical care -as it shows unbalance at 10% level (p-value = 0.06)-

4.3.3 Is this reduction in trustworthiness enough to compensate for the economic shock?

The negative shock generates an economic loss of 120 EP (60 EP per receiver). Therefore, it is intuitive that receivers in the treatment group have incentives to return less to compensate for this loss. Nevertheless, it is not intuitive if they are *over-compensating*, which may represent that they are taking advantage of the crisis, or if they are *under-compensating* and, consequently, they are not reimbursing enough to recover from the 120EP loss. Figure 8 depicts the receiver group's gains from the collective trust interaction. I denote the receiver group's gains as the difference between the **transferred amount** and the **returned amount**, which represents the amount of points that receivers kept from the trust interaction. In Figure 8, the total transferred amount is presented in color black (diamonds), the total returned amount (circles) is presented in solid blue (solid red) for control (treatment) group, and the receiver's gains (triangles) are depicted in dash blue (dash red) for control (treatment) group.

Figure 8: Receiver's gains from the Collective Trust Game

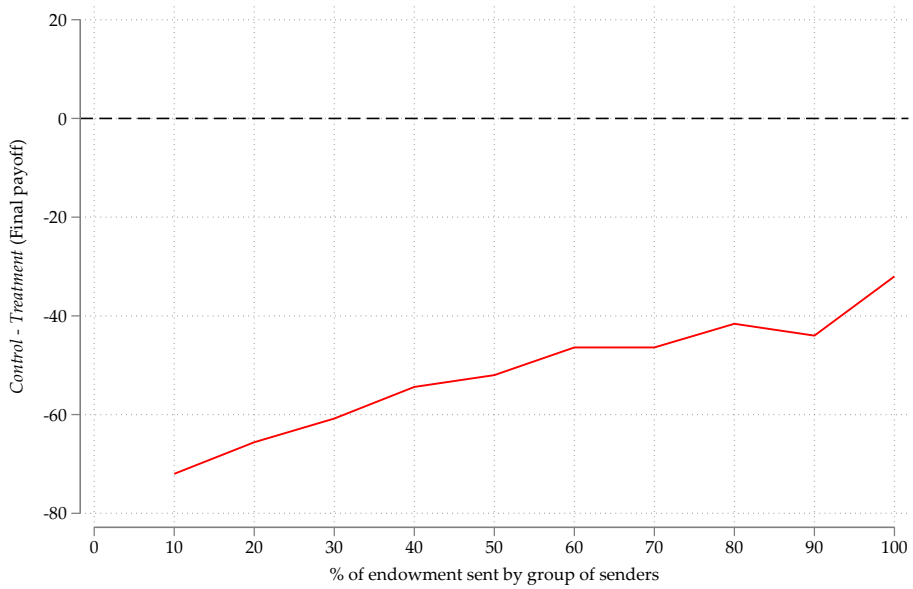


Note: Figure depicts in black (bar) the sent amount and black (diamonds) the total transfer amount for each possible transfer. e.i 10% transfer (20 EP sent amount from a 200 EP endowment) represents a 80 EP total transfer. 100% transfer (200 EP) represents an 800 EP total transfer. In solid lines (circles), the **points returned** represent the returned amount ($\approx$ 30%) for each possible transfer. Similarly, in dashed lines (triangles), the **points kept** represent the difference between *total transfer* and *returned amount*, which is close to 70%. Once the amount sent by senders increase the 20%, the experimental gains are always higher than 120 EP (which represents the experimental shock).

While the returned points (solid lines) are lower in the treatment group compared to the control group, the gains from the interaction are getting higher once the shock occurs (dashed lines). Furthermore, the gains from both groups once senders sent more than 20% of their endowment are always higher than 120EP, irrespective from the treatment status, which indicates that receivers are not aiming to compensate for the economic loss but instead are trying to recover the original gains. Moreover, consider the final payoff -which includes the experimental gains plus the initial endowment- between a control receiver $\pi_j(Control)$ and a treated receiver $\pi_j(Treated)$, and denote the gap between payoffs as the difference between both payments. Figure 9 shows this gap for each possible transferred amount.

Overall, Figure 9 shows that the difference in payoffs between a control receiver and treated receiver is decreasing as the group of senders increase their sent amount, ranging from an economic loss of 70 EP when senders sent 10% of their endowment, to 38 EP when senders sent 100% of theirs. These results indicate that receivers are not increasing their reliability and trustworthiness with the group of senders but instead are trying to recuperate their previous expected payoff via increasing their gains from the interaction of trust.

Figure 9: Difference in final payoffs between **receivers** in *Control* group vs *Treatment* group



Note: The solid red line denotes the difference in payoffs between a control receiver and a treated receiver across all possible sent amounts by the group of senders. The **payoff** captures the **endowment** + the experimental gains. Overall, the difference in payments is always lower than 120 EP (the experimental shock), which indicates that the experimental gains allow receivers to: (1) recover from the economic loss and (2) even try to recover their previous payoff. Finally, this reduction is higher once the total transfer increase.

5 Mechanisms

Now that I have shown that a negative economic shock caused a reduction in collective trust and trustworthiness, it is left to assess the mechanisms of these effects. Moreover, the importance of unpacking the mechanisms is higher given the differences that exist between the *homoaltruistic* model propositions in Section 2 and the empirical results depicted in Section 4.

The theoretical framework built in the model suggests that *inequality-averse* senders should increase their sent amount once the negative shock occurs. Yet, we observe a generalized reduction in the trust placed by senders, regardless of their distributional preferences. One plausible explanation for this result is the lack of inequality-averse subjects in the experimental sample, as students are commonly not inequality-averse agents (Kerschbamer and Müller, 2020; Epper et al., 2023), so their distributional preferences tend to reflect more a selfishness behavior.

In the following sections, I show that the decrease in collective trust is attributable to: (1) a decline in the sender's expected return amount from the group of receivers and (2) a persistent belief that the *other* sender will not reduce their sent amount, enabling the sender to free-ride on their partner's sent amount. Similarly, the decrease in collective trustworthiness is explained by: (1) a reduction in the receiver's expected returned amount by the *other* receiver, (2) a reduction in the receiver's expected sent amount by the group of senders, and (3) that receivers enforce what they perceive as the social norm (what should be done) once the shock occurs.

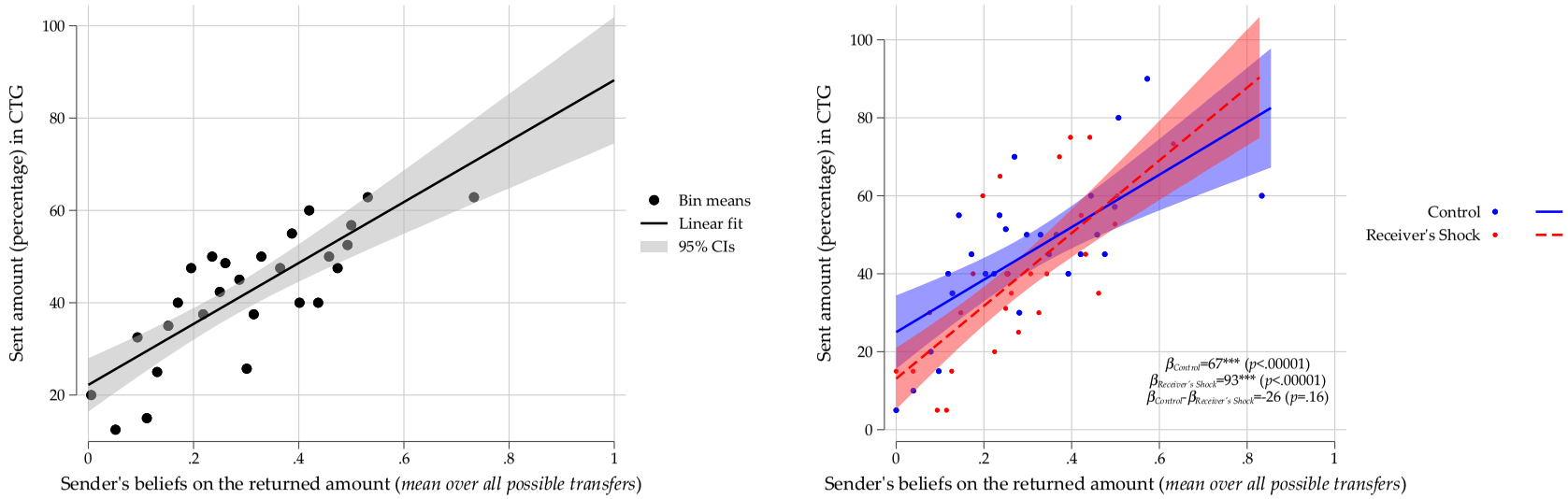
5.1 Why do we observe a reduction in trust?

The literature have consistently shown that trust is positively affected by the sender's expected trustworthiness (Ashraf et al., 2006; Kim et al., 2022). Hence, if we assume a positive relationship between collective trust and sender's first-order beliefs on the expected returned amount by receivers, then it is possible to disentangle the channel by exploring the effects of the economic shock over the sender's first-order beliefs.

5.1.1 Beliefs on the returned amount by the group of receivers

To validate this channel, I first analyze if the amount sent by the sender is correlated with his expectations of the returned amount (Kim et al., 2022). Given that I employ the strategy method to elicit the sender's first-order beliefs on the returned amount across all possible transfers, I first estimate the average across transfers and correlate that value with the sender's sent amount in the round before the shock. Figure 10a depicts a binned scatter plot between the sender's sent amount and the sender's beliefs on the returned amount.

Figure 10: **Trust** behavior across first-order beliefs on the returned amount (by round)



(a) Round *before* shock

(b) Round *after* shock

Note: Both figures depict binned scatter plots. **Left panel:** Relationship between trust behavior and expected trustworthiness is depicted for round before the shock. Both treatments were pooled as estimates in the round before shock are not statistically different. Each black dot represents the average sent amount (bin mean) for values at the sender's beliefs on the returned amount by the group of receivers. The linear fit (solid black) denotes the correlation between both variables. In light grey, 95% confidence interval for the linear estimation.

Right panel: Relationship between trust behavior and expected trustworthiness is depicted for round after the shock. The average sent amount (bin mean) by the control group is presented in blue. The average sent amount (bin mean) by the treatment group is presented in red. The linear fit (solid) denotes the correlation between both variables. In light blue, 95% confidence interval for the linear estimation for the control group. In light red, 95% confidence interval for the linear estimation for the treatment group.

From Figure 10a we observe a clear positive relationship between trust and expected trustworthiness (β = 0.6, p-value = 0.001), which is not unforeseen, as senders consistently show this behavior under relationships of trust (Barr, 2003). Furthermore, Figure 10b depicts the same relationship for the round after the shock disaggregated by treatment. Notice that the occurrence of the shock does not affect this positive relationship between treatments (p-value = 0.26), but it shows a general reduction in collective trust levels for low expected values in the returned amount.

Given that a decrease in the expected trustworthiness correlates with a decrease in trust, then if the economic shock is reducing the sender's first-order beliefs on the returned amount this may serve as a primary mechanism for the reduction in trust. To substantiate this statement, I propose the following model:

$$BelReturned_i^m = \alpha_0 + \alpha_1 Shock_i + x_{1i}'\beta + \varepsilon_i, \tag{12}$$

where $BelReturned_i^m$ denotes the sender's beliefs on the returned amount by the group of receivers for each possible sent amount m by his group.[21] Once again, $Shock_i$ indicates the treatment assignation, and (x_{1i}) denotes the set of controls for sender's behavior. The OLS estimates for each possible transfer are depicted in Table 7.

Similarly, the column 1 shows the sender's expected return amount (as percentage of total transfer) when his group sent 10% of the endowment, and column 10 shows the sender's expected return amount when his group sent 100%. Panel A shows that there are no differences between treatments in the expected returned average by senders in the round before the shock. It is worth mentioning that senders tend to expect a return amount close to the 33% of the transfer, which is not far from what is occurring (as receivers are returning 31%).

Regarding the treatment effect, as highlighted in Panel B, it is noteworthy that the economic shock leads to a reduction in the expected average return for low transfer values. For instance, senders anticipate a decrease of 11pp – equivalent to a 38% reduction in the perceived trustworthiness – when they transfer 10% of their endowment, and a decrease of 5pp when they transfer 20% of their endowment (p-value = 0.018). Notice that once the transferred amount increase from 20%, there are no differences in the expected returned amount between treatments. These results suggest that senders anticipate receivers utilizing scarce transfers to compensate for the economic losses, but once the transfer increase they expect receivers to be as trustworthy as they were before.

[21] Notice that the sender does not know the amount sent by his partner. Therefore, the strategy method allows me to recover his first-order beliefs about trustworthiness without providing additional information about the other sender.

Table 7: Sender beliefs on the returned amount by the group of receivers.

Dep Var: **Sender beliefs on the returned amount**	*Strategy Method*: Percentage of transfer by your group									
(as percentage of the total transferred)	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
Panel A: Round *before* shock										
Treatment	-0.02	0.00	-0.00	-0.01	-0.01	-0.01	-0.02	-0.03	-0.03	-0.01
	(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Constant	0.31**	0.31***	0.40***	0.45***	0.45***	0.36***	0.40***	0.44***	0.41***	0.40***
	(0.13)	(0.10)	(0.10)	(0.10)	(0.10)	(0.11)	(0.10)	(0.11)	(0.10)	(0.11)
Mean Dep. Variable	0.31	0.31	0.32	0.32	0.33	0.32	0.33	0.33	0.33	0.33
Effect size (%)	-6.3	1.5	-1.3	-3.6	-2.7	-2.8	-7.0	-8.1	-8.6	-3.5
Controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	238	238	238	238	238	238	238	238	238	238
Adjusted R-squared	0.05	0.11	0.08	0.11	0.09	0.13	0.12	0.11	0.12	0.08
Panel B: Round *after* shock	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
Treatment	-0.11***	-0.05**	-0.02	-0.03	-0.03	-0.04*	-0.03	-0.03	-0.04	-0.03
	(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Constant	0.58***	0.42***	0.41***	0.40***	0.50***	0.46***	0.47***	0.48***	0.45***	0.46***
	(0.11)	(0.10)	(0.10)	(0.11)	(0.09)	(0.09)	(0.10)	(0.10)	(0.10)	(0.10)
Mean Dep. Variable	0.29	0.30	0.30	0.31	0.32	0.32	0.32	0.32	0.33	0.33
Effect size (%)	-37.8	-16.5	-7.0	-10.9	-11.1	-13.3	-10.3	-9.5	-11.3	-9.5
Controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	238	238	238	238	238	238	238	238	238	238
Adjusted R-squared	0.16	0.14	0.09	0.10	0.13	0.13	0.14	0.14	0.13	0.14

Note: The set of controls included in ***Controls*** are: age, gender, stratum, risk measure, previous experience participating in experiments, and medical care -as it shows unbalance at 10% level (p-value = 0.06)-. The *Mean Dep. Variable* express the unconditional control mean of the dependent variable. The *Effect size (%)* denotes the size of the treatment effect as a percentage compared with the unconditional mean of the control group. Standard errors clustered at individual level in parenthesis. $^{***}p < 0.01$,$^{**}p < 0.05$,$^{*}p < 0.1$

Overall, the negative shock reduces collective trust through a decrease in the empirical expectations of the returned amount. This is evidenced by the negative effect of the shock over the sender's beliefs of trustworthiness and the positive relationship between trust and beliefs of collective trustworthiness shown in Figure 10a.

5.1.2 Beliefs on the sent amount by the *other* sender

Another potential channel for a reduction in trust is that the occurrence of a negative shock may lead senders to *free-ride* in other's contributions. To prove this channel, first recall that senders expect a reduction in trustworthiness (as shown above). Given this salient risk, if the sender believes that his partner would increase (or at least maintain) the amount sent, this could generate incentives to *free-ride* on his contribution.

To prove this channel, I first analyze the normative expectations about what senders consider to be *socially appropriate* to do after the shock. The left panel of Figure 11 depicts the distribution of the sender's response in the Krupka and Weber (2013) coordination game to the following inquiry:"After the shock, is it *socially appropriate* for the sender to send more, the same, or less than before?". As can be seen, nearly 73% of participants consider it socially appropriate to send **more** once the shock occurs. This result proves that (1) senders expect others to send more, as the social norm dictates that behavior and (2) despite the fact that senders consider it socially appropriate to send more after the sock, they are willing to "break" the social norm (Bogliacino et al., 2022).

Although senders consider it socially appropriate to increase (or at least maintain) the amount sent, it is not clear if that is what they expect from others. Thus, estimating the sender's empirical expectations about other's behavior is relevant. To do so, I estimate the effect of a negative economic shock over the sender's first-order beliefs using the following model:

$$BelSent_i^s = \alpha_0 + \alpha_1 Shock_i + x_{1i}'\beta + \varepsilon_i, \tag{13}$$

where $BelSent_i^s$ denotes the sender's first order belief on the amount sent by the *other* sender, $Shock_i$ indicates the treatment assignation, and (x_{1i}) denotes the set of controls for the sender's behavior. The OLS estimates are depicted in Table 8.

From columns 1 and 2, we observe that senders expect the other sender to transfer 43% of their endowment, and there are no statistically significant differences between treatments in the round before the shock (as expected). However, columns 3 and 4, which are the estimates once the shock occurs, reflect that expectations continue at 42%, which represents no variation in senders' beliefs about the amount sent by their counterparts.

This result suggests that senders expect their partner to keep sending 42 points, which represents

Table 8: Sender beliefs on the sent amount by his partner

Dep Var: **Sender beliefs on the sent amount by his partner**	Round			
	Before shock		After shock	
	(1)	(2)	(3)	(4)
Treatment	-2.65	-1.79	-2.16	-1.38
	(3.22)	(3.13)	(3.05)	(3.03)
Constant	43.67***	41.41***	42.33***	47.32***
	(2.23)	(13.31)	(2.12)	(12.85)
Mean Dep. Variable	43.7	43.7	42.3	42.3
Effect size (%)	-6.1	-4.1	-5.1	-3.3
Controls		✓		✓
Observations	238	238	238	238
Adjusted R-squared	0.00	0.07	0.00	0.11

Note: The set of controls included in ***Controls*** are: age, gender, stratum, risk measure, previous experience participating in experiments, and medical care -as it shows unbalance at 10% level (p-value = 0.06)-. The *Mean Dep. Variable* express the unconditional control mean of the dependent variable. The *Effect size (%)* denotes the size of the treatment effect as a percentage compared with the unconditional mean of the control group. Standard errors clustered at individual level in parenthesis. $^{***}p < 0.01$,$^{**}p < 0.05$,$^{*}p < 0.1$

20% of the total endowment shared by both senders (200 points). With this transfer, senders anticipate a 16% decline in the amount returned by the recipients (Column 2 of Table 7). In this scenario, senders may choose to send more resources in order to exceed the 20% criterion and observe normal levels of trustworthiness, but they are discouraged from sending more because of the possibility (salient risk of betrayal) that the other sender may not contribute as he did in the previous stage (Bohnet and Zeckhauser, 2004). In the aggregate, the negative shock undermines the entire equilibrium of trust previously established, increasing distrust toward the receivers and diminishing the trust placed by the senders.

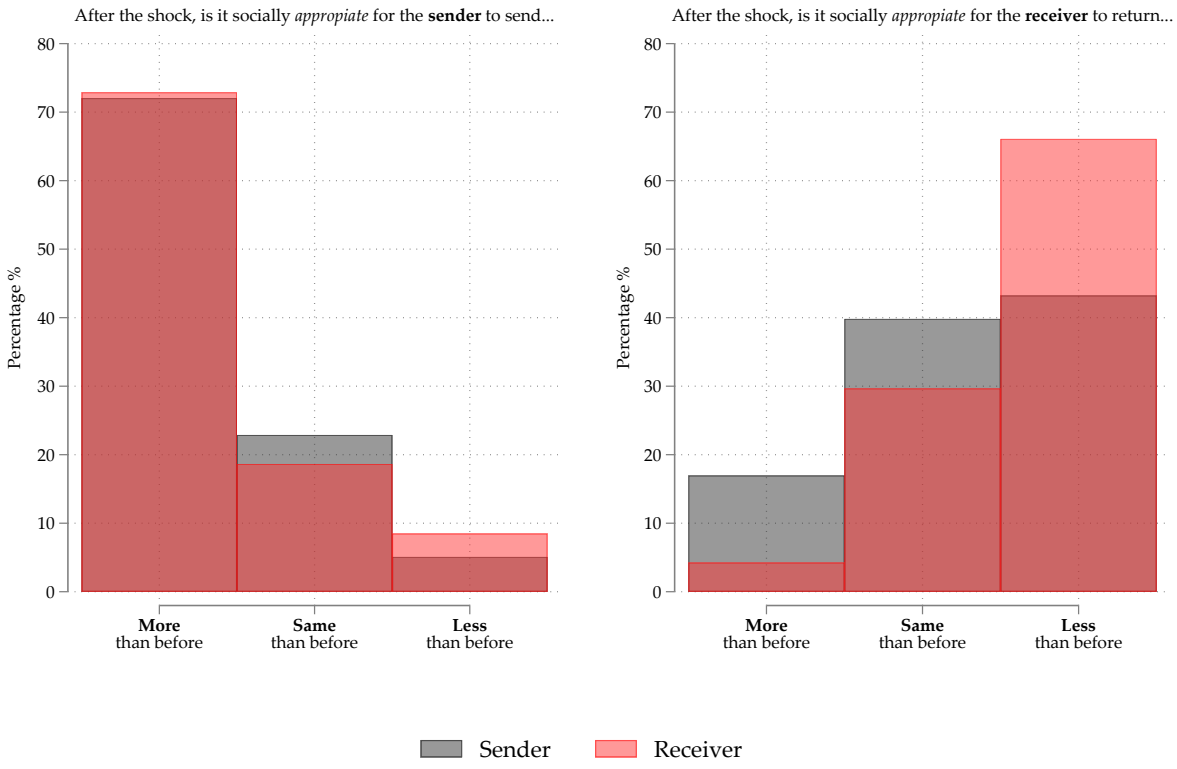
5.2 Why do we observe a reduction in trustworthiness?

Although trustworthiness is regarded as a well-established social norm (Bicchieri et al., 2011; Kimbrough and Vostroknutov, 2016), economic shocks may affect compliance with social norms (Bogliacino et al., 2022) or even change them. Therefore, the treatment may lead to a reduction in trustworthiness if: (1) receivers consider socially appropriate to return less than before, (2) receivers expect senders to reduce their sent amount, and (3) receivers expect other receivers to return less (Bicchieri and Xiao, 2009).

To explore the first channel, I undertake a comparative analysis of responses derived from the social norms questionnaire (Krupka and Weber, 2013) concerning the following inquiry: "After the shock, is it *socially appropriate* for the Receiver to return more, the same, or less than before?". In this questionnaire, both types of participants provided responses trying to match the appropriateness

rate held by *another* participant of the same type. Right panel of Figure 11 illustrates the distribution of these responses, with responses from senders depicted in black and those from receivers in red.

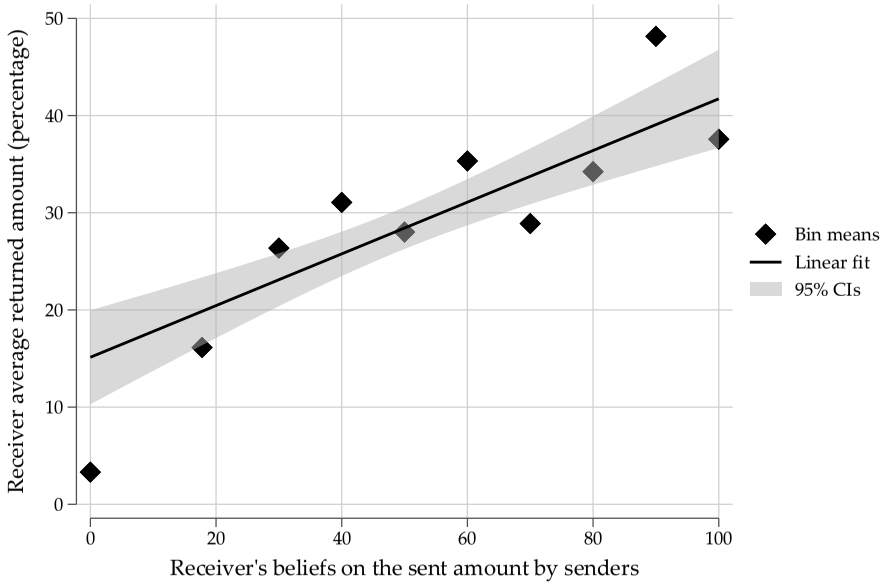
Figure 11: Social norm after the shock



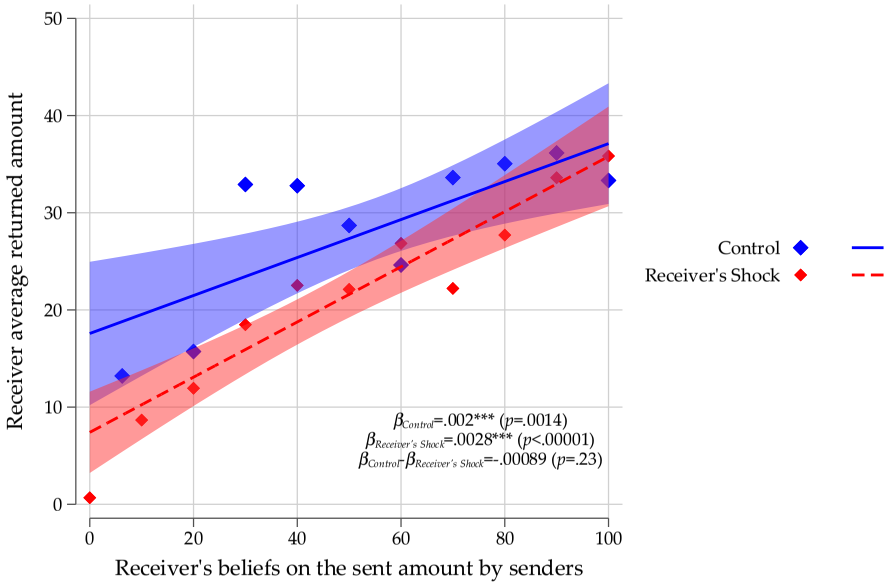
As depicted in the left panel, following an economic shock that diminishes the endowment of receivers, the social norm (descriptive norm) regarding trust is to: transfer **more** resources than before, with no discernible disparities between the opinions of both participant types (χ^2 test p-value = 0.469). In the right panel, regarding trustworthiness the social norm dictates: returning **less** than before. Furthermore, the distribution of responses indicates that the action of "returning less" is considerably more preferred and endorsed by receivers (66% of them) than by senders (43% of them) (χ^2 test p-value = 0.000), demonstrating that receivers perceive it socially appropriate to return less to compensate for the shock. Finally, Figure D7 and Figure D8 in the Appendix depicts potential differences between participant types in the normative expectations about trust and trustworthiness in a general framework.[22]

[22] Participants were asked to rate the socially appropriateness for the sent amount and returned amount in the round *before the shock* where all participants had the same endowment of 100 EP.

Figure 12: **Trustworthiness** behavior across first-order beliefs on the sent amount (by round)



(a) Round *before* shock



(b) Round *after* shock

Note: Both figures depict binned scatter plots. **Left panel:** Relationship between trustworthiness behavior and expected trust is depicted for round before the shock. Both treatments were pooled as estimates in the round before shock are not statistically different. Each black diamond represents the average returned amount (bin mean) for values at the receiver's beliefs on the sent amount by the group of senders. The linear fit (solid black) denotes the correlation between both variables. In light grey, 95% confidence interval for the linear estimation.

Right panel: Relationship between trustworthiness behavior and expected trust is depicted for round after the shock. The average returned amount (bin mean) by the control group is presented in blue. The average returned amount (bin mean) by the treatment group is presented in red. The linear fit (solid) denotes the correlation between both variables. In light blue, 95% confidence interval for the linear estimation for the control group. In light red, 95% confidence interval for the linear estimation for the treatment group.

5.2.1 Beliefs on the sent amount by the group of senders

Similar to trust, trustworthiness can also depend on beliefs about the trust placed. This phenomenon arises from the tendency of individuals to be trustworthy when they are expected to be so (Reuben et al., 2009). In order to verify this channel, I first examine if the receiver's transfer amount and his expectations of the sent amount are correlated. I employ the average returned amount across transfers and correlate that value with the receiver's expected sent amount in the round before the shock. Figure 12a shows a binned scatter plot between these two variables.

As can be seen in Figure 12a, we observe a strong positive correlation between expected trust and trustworthiness (β = 0.0027, p-value = 0.001). Additionally, Figure 12b depicts this relationship for the round after the shock. Notice that the occurrence of the shock does not affect this positive relationship between treatments (p-value = 0.26), but it shows a general reduction in trustworthiness levels for low expected values in the sent amount.

Since a decline in the receiver's expected sent amount is correlated with a decline in trustworthiness, the primary explanation for the decline in trustworthiness may be that the economic shock is reducing the receiver's first-order beliefs about the sent amount. To empirically assess this mechanism, I estimate the following model:

$$BelSent_j^r = \alpha_0 + \alpha_1 Shock_j + x_{1j}'\beta + \varepsilon_j, \tag{14}$$

where $BelSent_j^r$ denotes the receiver's first order belief on the amount sent by the group of senders, $Shock_i$ indicates the treatment assignment, and (x_{1j}) denotes the set of controls for the receiver's behavior. The OLS estimates are depicted in Table 9.

As observed in columns 1 and 2 from Table 9, there are no differences between treatments in the expected sent amount in the round before the shock. Receivers expect on average a 53% transfer (SD = 0.26) while senders are transferring 43% (see Table 2) being this difference statistically significant (p-value = 0.000). In general, this result indicates that receiver's expectations are not aligned with empirical values as receivers expect senders to trust more than what they do.

Regarding the effects of the shock, a 60% reduction in the receiver's endowment leads to a $7pp$ reduction in the receiver's expected sent amount. Furthermore, coefficients are notably large as they range between a -11% to -15% reduction compared to the control's mean. Overall, receivers expect senders to mistrust them by sending less, which leads to a reduction in the returned amount. Figure 12b depicts this result.

Table 9: Receiver beliefs on the sent amount by the group of the senders

Dep Var: **Receiver beliefs on the sent amount by senders**	Round			
	Before shock		After shock	
	(1)	(2)	(3)	(4)
Treatment	-1.80	-1.98	-6.43*	-7.42**
	(3.38)	(3.23)	(3.51)	(3.37)
Constant	53.58***	41.12***	55.92***	44.29***
	(2.35)	(13.19)	(2.41)	(14.18)
Mean Dep. Variable	53.6	53.6	55.9	55.9
Effect size (%)	-3.4	-3.7	-11.5	-13.3
Controls		✓		✓
Observations	238	238	238	238
Adjusted R-squared	0.00	0.12	0.01	0.11

Note: The set of controls included in ***Controls*** are: age, gender, stratum, risk measure, previous experience participating in experiments, and medical care -as it shows unbalance at 10% level (p-value = 0.06)-. Robust standard errors in parenthesis. $^{***}p < 0.01, ^{**}p < 0.05, ^{*}p < 0.1$

5.2.2 Beliefs on the returned amount by the *other* receiver

Finally, if we consider trustworthiness as a signal for pro-social behavior (Fehrler, 2010), then it is possible that lower beliefs in the trustworthiness of others may reduce the incentives to behave in a trustworthy manner (Frey and Meier, 2004). Therefore, if the economic shock motivates a reduction in the receiver's expected returned amount by the other receiver, then this could represent another channel for the reduction in collective trustworthiness.

To empirically assess this channel, I estimate the following model:

$$BelReturned_j^r = \alpha_0 + \alpha_1 Shock_i + x_{1j}'\beta + \varepsilon_j, \tag{15}$$

where $BelReturned_j^r$ denotes the receiver's beliefs on the returned amount by the other receiver. $Shock_i$ indicates the treatment assignation, and (x_{1j}) denotes a set of controls for receiver's behavior. The OLS estimates for each possible transfer are depicted in Table 10. Column 1 depicts the receiver's expected return amount (as percentage of total transfer) when senders sent 10% of the endowment, and column 10 shows the receiver's expected return amount when senders sent 100%.

Panel A shows that there are no differences between treatments in the expected returned average in the round before shock, and receivers expect others to return 30% of the transfer, which is close to what they are actually returning (observe Table 5).[23]

[23] This result may indicate the presence of consensus effect (Adriani and Sonderegger, 2015), although research about this type of behavior may be necessary to establish this relationship.

Table 10: Receiver beliefs on the returned amount by his partner

Dep Var: **Receiver beliefs on the returned**	*Strategy Method*: Percentage of transfer by the group of senders									
amount by his partner *(as percentage of the received transferred)*	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
Panel A: Round *before* shock										
Treatment	0.00	0.02	-0.01	-0.02	0.01	-0.00	-0.00	-0.01	-0.01	-0.00
	(0.03)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Constant	0.10	0.12	0.09	0.13	0.05	0.06	0.07	0.05	0.05	0.06
	(0.15)	(0.11)	(0.11)	(0.10)	(0.10)	(0.10)	(0.10)	(0.10)	(0.11)	(0.10)
Mean Dep. Variable	0.30	0.30	0.32	0.32	0.32	0.31	0.31	0.31	0.31	0.31
Effect size (%)	1.3	5.5	-3.8	-4.8	2.1	-1.5	-0.3	-3.0	-2.1	-1.6
Controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	238	238	238	238	238	238	238	238	238	238
Adjusted R-squared	0.09	0.09	0.11	0.10	0.08	0.09	0.08	0.06	0.07	0.06
Panel B: Round *after* shock	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
Treatment	-0.11***	-0.07***	-0.06***	-0.09***	-0.08***	-0.07***	-0.06***	-0.07***	-0.06**	-0.06***
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Constant	0.25**	0.13	0.12	0.09	0.14	0.16	0.17	0.13	0.08	0.13
	(0.10)	(0.10)	(0.10)	(0.10)	(0.10)	(0.10)	(0.10)	(0.11)	(0.10)	(0.10)
Mean Dep. Variable	0.30	0.29	0.31	0.32	0.33	0.32	0.32	0.32	0.31	0.31
Effect size (%)	-35.0	-23.1	-20.5	-27.0	-24.1	-21.2	-18.5	-20.5	-17.9	-19.8
Controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	238	238	238	238	238	238	238	238	238	238
Adjusted R-squared	0.19	0.12	0.13	0.14	0.13	0.13	0.12	0.12	0.10	0.10

Note: The set of controls included in ***Controls*** are: age, gender, stratum, risk measure, previous experience participating in experiments, and medical care -as it shows unbalance at 10% level (p-value = 0.06)-. The *Mean Dep. Variable* express the unconditional control mean of the dependent variable. The *Effect size (%)* denotes the size of the treatment effect as a percentage compared with the unconditional mean of the control group. Standard errors clustered at individual level in parenthesis. $^{***}p < 0.01,^{**}p < 0.05,^{*}p < 0.1$

Regarding the treatment effect, as highlighted in Panel B of Table 10, the economic shock leads to a reduction in the expected average return amount for all possible transfers. For instance, receivers anticipate a decrease of 11*pp* – equivalent to a 35% reduction in trustworthiness by the other receiver when senders transfer 10% of their endowment, and a decrease of 6*pp* when they transfer 100%. Moreover, receiver's expect a higher reduction in trustworthiness compared with the actual reduction, being the latter 2*pp* lower in average (See Table 5). Finally, note that receivers expect a more significant reduction in the amount returned compared to the expectations of senders. This result aligns with the differences in the social norm discussed above, demonstrating that receivers not only consider it normative to return less, but empirically expect their partner to return less. In the aggregate, these two channels incentivize the receiver to return less after the shock.

5.2.3 Heterogeneous effects

Finally, I explore for heterogeneous treatment effects of the negative economic shock by political preferences and personal attitudes. Given that the theoretical model presents some insights on how prosocial behavior affects trust, this analysis provides useful information to confirm the theoretical findings, and explore how personal and political views affect the response to economic shocks. To estimate these effects, I propose the following model:

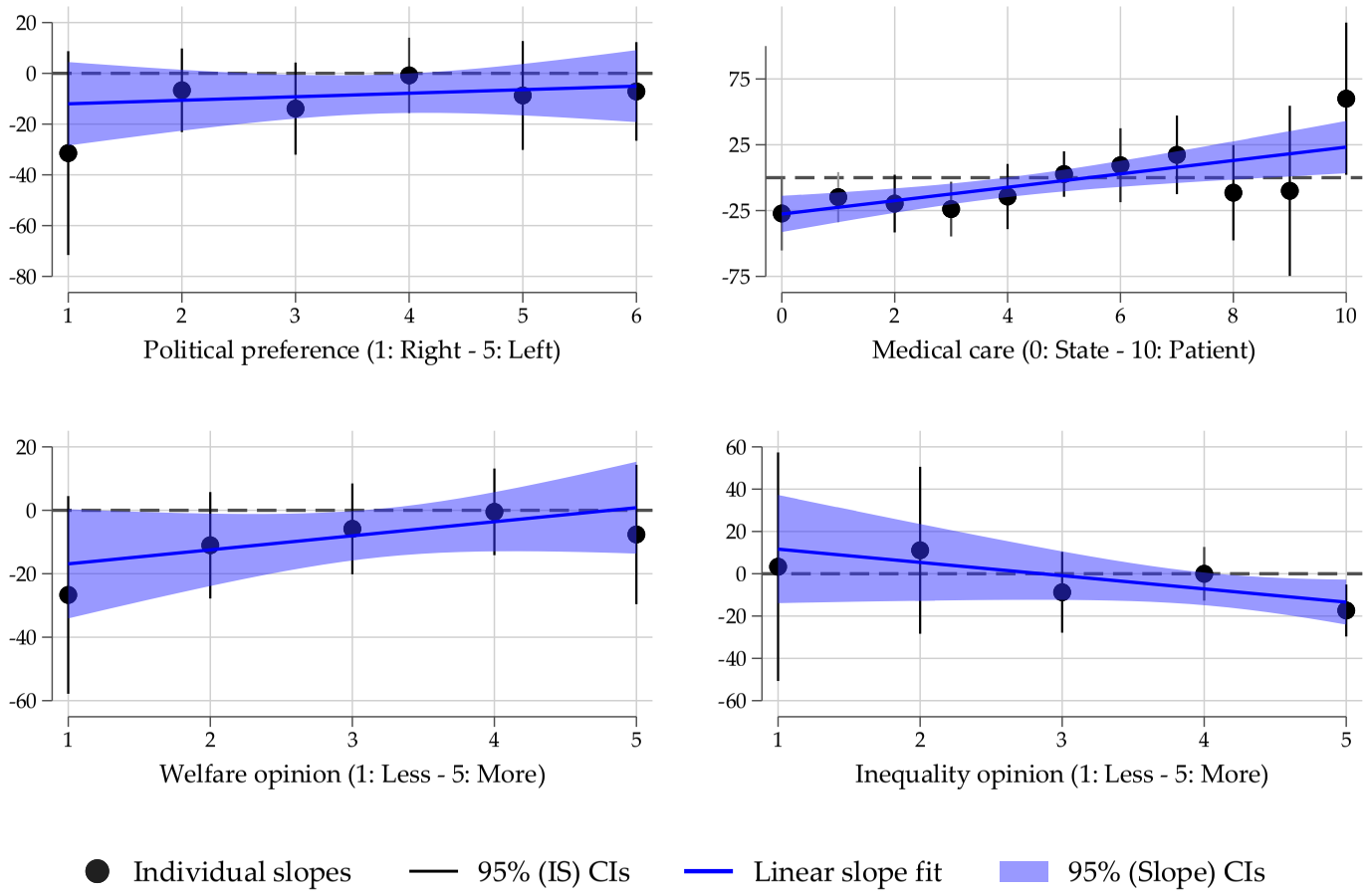
$$Y_i = \beta_0 + \beta_1 Shock_i + \beta_2 Z_i + \gamma Shock_i \times Z_i + \varepsilon_i \tag{16}$$

where Y_i denotes the outcomes of interest, $Shock_i$ indicates the treatment assignation and Z_i denotes a variable that captures a political preference or an attitude. For this analysis, I employ as Z_i variables the participant's answers to the following questions:

- (Z_1) When you think about your political viewpoint, would you describe yourself as right-wing (1), center-right (2), center (3), center-left (4), left-wing (5), does not know (6)?
- (Z_2) If a person gets sick, who should pay for their treatment? (0: State - 10: Patient)
- (Z_3) How much do you agree with the following statement: Each individual is responsible for their own well-being (1: Totally disagree - 5: Totally agree)
- (Z_4) How much do you agree with the following statement: The government should implement strong policies to reduce inequality between the rich and the poor (1: Totally disagree - 5: Totally agree).

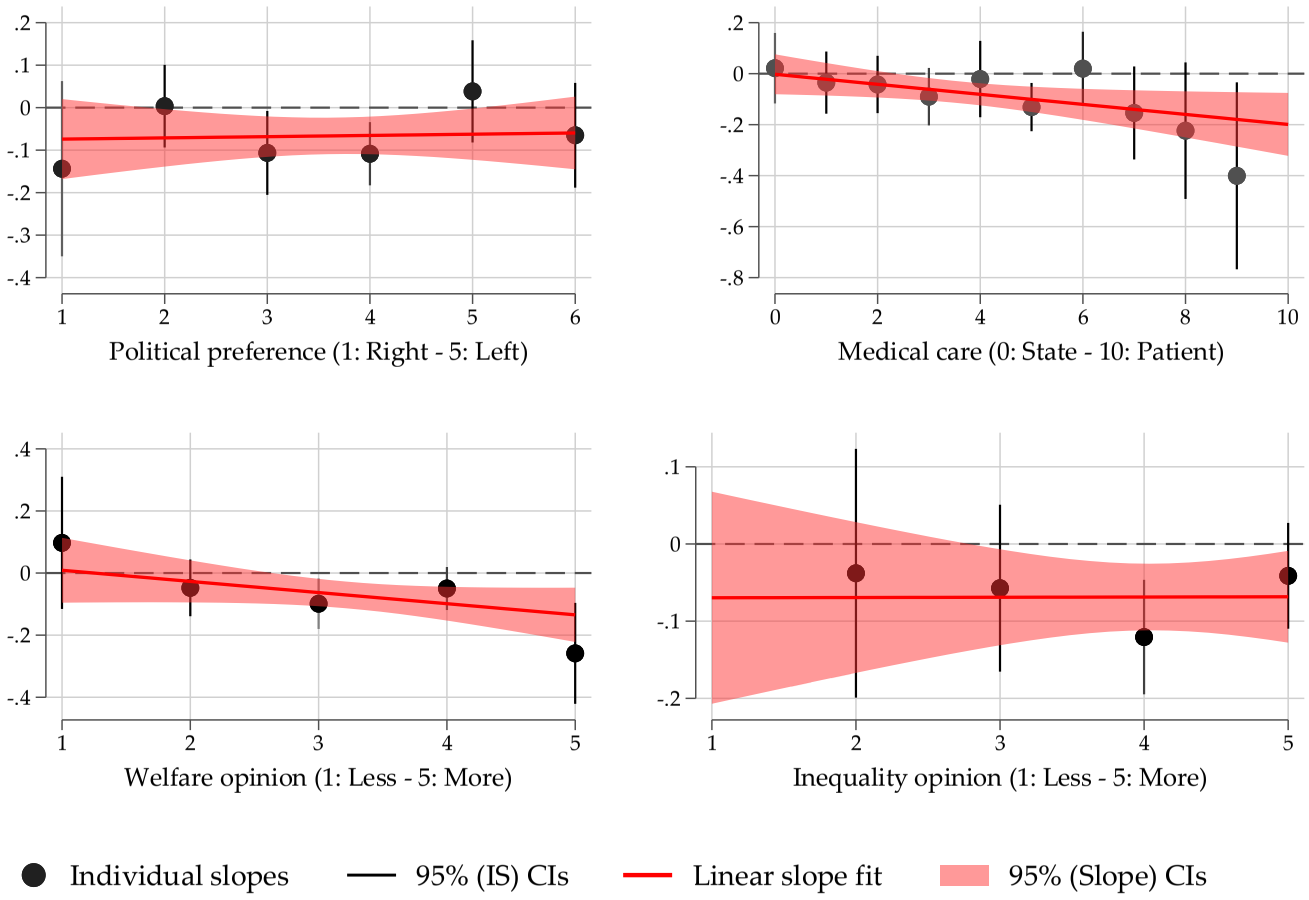
Overall, the average participant considers that the state should pay more than the patient for a medical treatment (Mean = 3.73, SD = 2.4), that the government should implement firm policies to reduce inequality (Mean = 4.05, SD = 0.9) and that each individual is responsible for their own well-being (Mean = 3.22, SD = 1). The distribution of responses for questions (Z_i) are depicted in Figure D6 in the Appendix.

Figure 13: Heterogeneous treatment effect over **Collective Trust** across attitudes and preferences.



Note: The individual slopes (circles) depicts the treatment effect for a sub sample of subjects with $Z_i = Z^l$ in the round after the shock. In black, 95% confidence intervals for the individual slopes (IS). The conditional marginal treatment effect is presented in blue. The 95% confidence interval for the linear trend is presented in lighted blue. **Political preference**: denotes the participant's answer to the following question: When you think about your political viewpoint, would you describe yourself as right-wing (1), center-right (2), center (3), center-left (4), left-wing (5), does not know (6)? **Medical care**: is a variable that indicates between 0 (State) and 10 (Patient) who should paid more for a medical treatment. **Welfare opinion**: is a categorical variable that ranges from 1 = "Totally disagree" to 5 = "Totally agree" to the statement: "Each individual is responsible for their own well-being". **Inequality opinion** is a categorical variable that ranges from 1 = "Totally disagree" to 5 = "Totally agree" to the statement: "The government should implement strong policies to reduce inequality".

Figure 14: Heterogeneous treatment effect over **Collective Trustworthiness** across attitudes and preferences.



Note: The individual slopes (circles) depicts the treatment effect for a sub sample of subjects with $Z_i = Z^l$ in the round after the shock. In black, 95% confidence intervals for the individual slopes (IS). The conditional marginal treatment effect is presented in red. The 95% confidence interval for the linear trend is presented in lighted red. **Political preference**: denotes the participant's answer to the following question: When you think about your political viewpoint, would you describe yourself as right-wing (1), center-right (2), center (3), center-left (4), left-wing (5), does not know (6)? **Medical care**: is a variable that indicates between 0 (State) and 10 (Patient) who should paid more for a medical treatment. **Welfare opinion**: is a categorical variable that ranges from 1 = "Totally disagree" to 5 = "Totally agree" to the statement: "Each individual is responsible for their own well-being". **Inequality opinion** is a categorical variable that ranges from 1 = "Totally disagree" to 5 = "Totally agree" to the statement: "The government should implement strong policies to reduce inequality".

Figure 13 and 14 display the conditional marginal treatment effect for each Z_i variable. In particular, each dot (individual slope) represents the estimate of the treatment effect for a particular value of Z_i, which is obtained by estimating the following equation:

$$Y_i = \beta_0 + \beta_1 Shock_i + \varepsilon_i \quad \text{if} \quad Z_i = Z^l$$

Additionally, the linear slope (solid line) represents the conditional marginal effect (γ) estimated in Equation (16). Finally, the OLS estimates for trust and trustworthiness behavior of Equation (16) are presented in Table D3 and Table D4 in the Appendix.

From Figure 13 it is possible to document interesting patterns. First, concerning political preferences effects, I find no difference in the treatment effect over collective trust across political viewpoints. However, I find differences across political opinions. For instance, notice that people who support that the state should be responsible for a medical treatment are more likely to reduce the sent amount in the round after shock, while people that do not consider inequality to be an important concern are more likely to send the same amount than before. These results may indicate that individuals more aligned with the idea of a *welfare state* are less likely to trust more after shocks, while more *individualistic* subjects are less likely to be affected by the shock, but further research is necessary.

Finally, regarding heterogeneous treatment effects over collective trustworthiness observe Figure 14. Notice that people who support that the patient should pay for the medical treatment, or consider that each individual is responsible of their own well-being are more likely to reduce the returned amount. This result aligns with less prosocial agents being more likely to reduce the amount they return (Cardenas et al., 2013), and being more likely to be affected by negative economic shocks.

6 Conclusions

This paper has studied the effect of negative economic shocks over collective trust and collective trustworthiness. While prosocial theoretical models (Fehr and Schmidt, 1999; Bolton and Ockenfels, 2000) predict a positive impact of economic shocks over trust, a laboratory experiment has shown that negative economic shocks have harmful impacts on collective trust and collective trustworthiness. Moreover, I find robust evidence that participants exposed to negative economic shocks tend to trust less and are more likely to reduce the amount sent in a modified collective trust game (Berg et al., 1995; Holm and Nystedt, 2010). Regarding trustworthiness behavior, I find that negative economic shocks also decrease collective trustworthiness, showing that people affected by the shock reduce their returned amount between -11*pp* to -4*pp*, depending on the amount transferred by the group of senders. Moreover, these reductions in trustworthiness are economically significant and range between 30% to 14% compared to the control group mean.

To delve into potential mechanisms behind these results, I assess the impact of economic shocks on first-order beliefs and normative views regarding socially appropriate behavior. Consistent with existing literature, this study highlights empirical expectations and normative social norms as pivotal factors driving this decline in collective trust. Beliefs play a fundamental role in trust relationships, particularly in group trust dynamics. Group relationships pave the way for collective action problems and coordination issues. Therefore, the impacts of economic shocks on beliefs about others' behavior have significant negative effects and translate into a general erosion of group trust.

Overall, this study contributes to the understanding of group trust dynamics under exogenous income shocks, and results pose avenues for further research. For instance, following papers should aim to formalize the theoretical rationality behind these reductions in collective trust and trustworthiness, considering strategic behavior from both trustors and trustees. Furthermore, there is a pressing need to delve deeper into the role of distributional preferences, such as altruism, selfishness, or spitefulness, in explaining these adverse effects. By directing attention towards these aspects, future studies can elucidate the mechanisms through which economic shocks impact collective trust and how we can recover from these harmful impacts.

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Appendices

A Payoff functions

A.1 Sender's payoff

Suppose a two (2) senders and two (2) receivers scenario. As mentioned in Section 2, each sender obtains the following material payoff:

$$\pi_1^s = E_1^s + \frac{1}{2}\left[\gamma_1^r\left(\frac{\kappa}{2}(M_1^s + M_2^s)\right) + \gamma_2^r\left(\frac{\kappa}{2}(M_1^s + M_2^s)\right)\right] - M_1^s$$

$$\pi_2^s = E_2^s + \frac{1}{2}\left[\gamma_1^r\left(\frac{\kappa}{2}(M_1^s + M_2^s)\right) + \gamma_2^r\left(\frac{\kappa}{2}(M_1^s + M_2^s)\right)\right] - M_2^s$$

Here, the first term denotes the initial endowment. Inside the brackets is the total transferred amount returned by receivers, which is a function of M_1 and M_2. The total returned amount is equally distributed, among both receivers ($\frac{1}{2}$). Finally, the last term M_1 denotes that each token transferred to receivers supposes a cost for the sender. We obtain the following payoff by extending this function for a more general framework:

$$\pi_i^s = E_i^s + \frac{1}{S}\left[\frac{\kappa}{R}\left(\sum_{l=1}^{S} M_l^s\right)\left(\sum_{j=1}^{R}\gamma_j^r\right)\right] - M_i^s$$

$$\pi_i^s = E_i^s + \frac{1}{S}\left[\kappa\left(\sum_{l=1}^{S} M_l^s\right)(\overline{\gamma^r})\right] - M_i^s$$

$$\pi_i^s = E_i^s + \frac{\kappa}{S}\left[(\overline{\gamma^r})\left(M_i^s + \sum_{m\neq i}^{S} M_m^s\right)\right] - M_i^s$$

$$\pi_i^s(M_i^s) = E_i^s + M_i^s\left(\frac{\kappa}{S}(\overline{\gamma^r}) - 1\right) + \underbrace{\frac{\kappa}{S}\left[(\overline{\gamma^r})\left(\sum_{m\neq i}^{S} M_m^s\right)\right]}_{Exogenous} \tag{17}$$

A.2 Receiver's payoff

As mentioned in Section 2, receiver obtains the following material payoff:

$$\pi_1^r = E_1^r + \left(\frac{\kappa}{2}(M_1^s + M_2^s)\right)(1 - \gamma_1^r)$$

$$\pi_2^r = E_2^r + \left(\frac{\kappa}{2}(M_1^s + M_2^s)\right)(1 - \gamma_2^r)$$

In this first scenario, M_1 denotes the transferred amount sent by Sender 1 and M_2 denotes the transferred amount sent by Sender 2. Additionally, we have $S = 2$ and $R = 2$. For a general scenario,

we extend the receivers payoff to be:

$$\pi_j^r(M_i^s) = E_j^r + \left(\frac{\kappa}{R}\sum_{i=1}^{S} M_i^s\right)(1-\gamma_j^r). \tag{18}$$

A.3 Average receiver's payoff

The average receiver's payoff is defined as:

$$\overline{\pi_j^r}(M_i^s) = \left(\sum_{j=1}^{R}\frac{\pi_j^r(M_i^s)}{R}\right).$$

Using Equation (18) and rearranging some terms, we obtain

$$\begin{aligned}
&= \frac{1}{R}\sum_{j=1}^{R}\left(E_j^r + \left(\frac{\kappa}{R}\sum_{i=1}^{S} M_i^s\right)(1-\gamma_j^r)\right) \\
&= \overline{E_j^r} + \left(\frac{\kappa}{R}\sum_{i=1}^{S} M_i^s\right)\left(\frac{1}{R}\sum_{j=1}^{R}(1-\gamma_j^r)\right) \\
&= \overline{E_j^r} + \left(\frac{\kappa}{R}\sum_{i=1}^{S} M_i^s\right)\left(\frac{1}{R}\sum_{j=1}^{R}1 - \frac{1}{R}\sum_{j=1}^{R}\gamma_j^r\right)
\end{aligned}$$

$$\boxed{\overline{\pi_j^r}(M_i^s) = \overline{E_j^r} + \left(\frac{\kappa}{R}\sum_{i=1}^{S} M_i^s\right)(1-\overline{\gamma_j^r}).} \tag{19}$$

B Theoretical Homo Altruistus Equilibrium

From Equation (3) we know that an inequality-averse sender is maximizing:

$$\max_{M_i^s} U_i(M_i^s) = \pi_i^s(M_i^s) - \frac{\alpha}{2}(\pi_i^s(M_i^s) - \overline{\pi_j^r}(M_i^s))^2.$$

Let me rewrite the maximization problem using Equation (17) and Equation (19), and we obtain

$$\max_{M_i^s} U_i = \underbrace{E_i^s + M_i^s\left(\frac{\kappa}{S}(\overline{\gamma^r}) - 1\right) + \frac{\kappa}{S}\left[(\overline{\gamma^r})\left(\sum_{m\neq i}^{S} M_m^s\right)\right]}_{\pi_i^s} -$$

$$\frac{\alpha}{2}\left(\underbrace{E_i^s + M_i^s\left(\frac{\kappa}{S}(\overline{\gamma^r}) - 1\right) + \frac{\kappa}{S}\left[(\overline{\gamma^r})\left(\sum_{m\neq i}^{S} M_m^s\right)\right]}_{\pi_i^s} - \underbrace{\left(\overline{E_j^r} + \left(\frac{\kappa}{R}\sum_{i=1}^{S} M_i^s\right)(1-\overline{\gamma_j^r})\right)}_{\pi_j^r}\right)^2 \tag{20}$$

For facility in notation, I define $\mu = \left(\frac{\kappa\overline{\gamma_r}}{S}\right)$, $\lambda = \left(\sum_{m\neq i}^{S} M_m^s\right)$, $\sigma = \left(\frac{\kappa(1-\overline{\gamma_r})}{R}\right)$, and then rearrange some terms in Equation (20)

$$\max_{M_i^s} E_i^s + M_i^s(\mu-1) + (\mu\lambda) - \frac{\alpha}{2}\left(E_i^s + M_i^s(\mu-1) + (\mu\lambda) - \left(\overline{E_j^r} + \left(\frac{\kappa}{R}\sum_{i=1}^{S} M_i^s\right)(1-\overline{\gamma_j^r})\right)\right)^2$$

$$\max_{M_i^s} E_i^s + M_i^s(\mu-1) + (\mu\lambda) - \frac{\alpha}{2}\left(E_i^s + M_i^s(\mu-1) + (\mu\lambda) - \left(\overline{E_j^r} + \frac{\kappa(1-\overline{\gamma_j^r})}{R}\left(M_i^s + \sum_{m\neq i}^{S} M_m^s\right)\right)\right)^2$$

$$\max_{M_i^s} E_i^s + M_i^s(\mu-1) + (\mu\lambda) - \frac{\alpha}{2}\left(E_i^s + M_i^s(\mu-1) + (\mu\lambda) - \left(\overline{E_j^r} + \frac{M_i^s\kappa(1-\overline{\gamma_j^r})}{R} + \frac{(\sum_{m\neq i}^{s} M_m^s)\kappa(1-\overline{\gamma_j^r})}{R}\right)\right)^2$$

Simplifying notation,

$$\max_{M_i^s} \pi_i^s = E_i^s + M_i^s(\mu-1) + (\mu\lambda) - \frac{\alpha}{2}\left(E_i^s + M_i^s(\mu-1) + (\mu\lambda) - \overline{E_j^r} - M_i^s\sigma - \lambda\sigma\right)^2.$$

Since the objective function is continuous, concave in M_i and twice differentiable, it is possible to find the maximum of the function by applying the first order conditions:

$$\frac{\partial \pi_i^s}{\partial M_i^s} = (\mu-1) - \alpha\left(E_i^s + M_i^s(\mu-1) + (\mu\lambda) - \overline{E_j^r} - M_i^s\sigma - \lambda\sigma\right)(\mu-1-\sigma) = 0,$$

which rearranging yields in

$$\mu - 1 - \alpha(E_i^s - \overline{E_j^r})(\mu-1-\sigma) - \alpha M_i^s(\mu-1-\sigma)^2 - \alpha\lambda(\mu-\sigma)(\mu-1-\sigma) = 0$$

$$\alpha M_i^s(\mu-1-\sigma)^2 = \mu - 1 - \alpha(E_i^s - \overline{E_j^r})(\mu-1-\sigma) - \alpha\lambda(\mu-\sigma)(\mu-1-\sigma)$$

Solving for M_i^s, we obtain the following expression:

$$\boxed{M_i^{s^*} = \frac{\frac{\kappa\overline{\gamma_r}}{S} - 1}{\alpha\left(\frac{\kappa\overline{\gamma_r}}{S} - \frac{\kappa(1-\overline{\gamma_r})}{R} - 1\right)^2} - \frac{E_i^s - \overline{E_j^r}}{\left(\frac{\kappa\overline{\gamma_r}}{S} - \frac{\kappa(1-\overline{\gamma_r})}{R} - 1\right)} - \frac{\left(\sum_{m\neq i}^{S} M_m^s\right)\left(\frac{\kappa\overline{\gamma_r}}{S} - \frac{\kappa(1-\overline{\gamma_r})}{R}\right)}{\left(\frac{\kappa\overline{\gamma_r}}{S} - \frac{\kappa(1-\overline{\gamma_r})}{R} - 1\right)}}$$

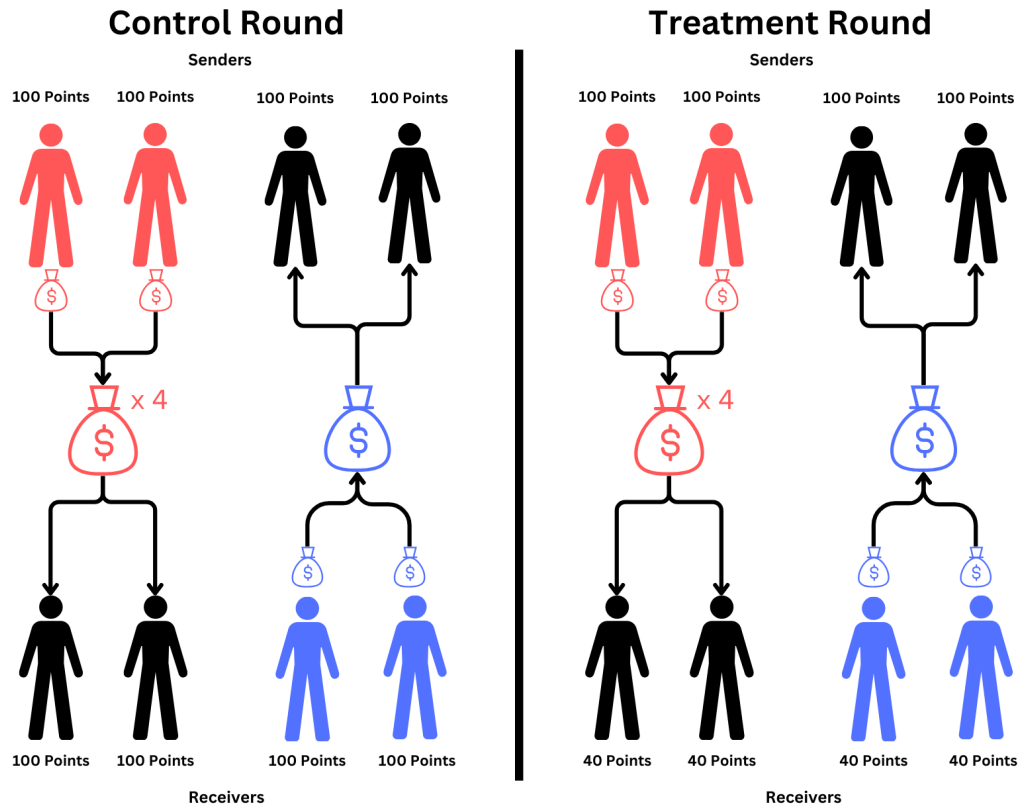
C Experimental Details

Table C1: Endowment's across participants, rounds and treatments.

	Round	
Sender's Endowment	*Before shock*	*After shock*
Control Group	100 EP	100 EP
Treatment Group	100 EP	100 EP
Receiver's Endowment	*Before shock*	*After shock*
Control Group	100 EP	100 EP
Treatment Group	100 EP	40 EP

Note: Participants were assigned to one role, and this role was fixed during the entire experiment. 100 EP = 20000 COP (5 USD). Treatment effect can be seen in red, where receiver's suffer a 60% reduction in their initial endowment.

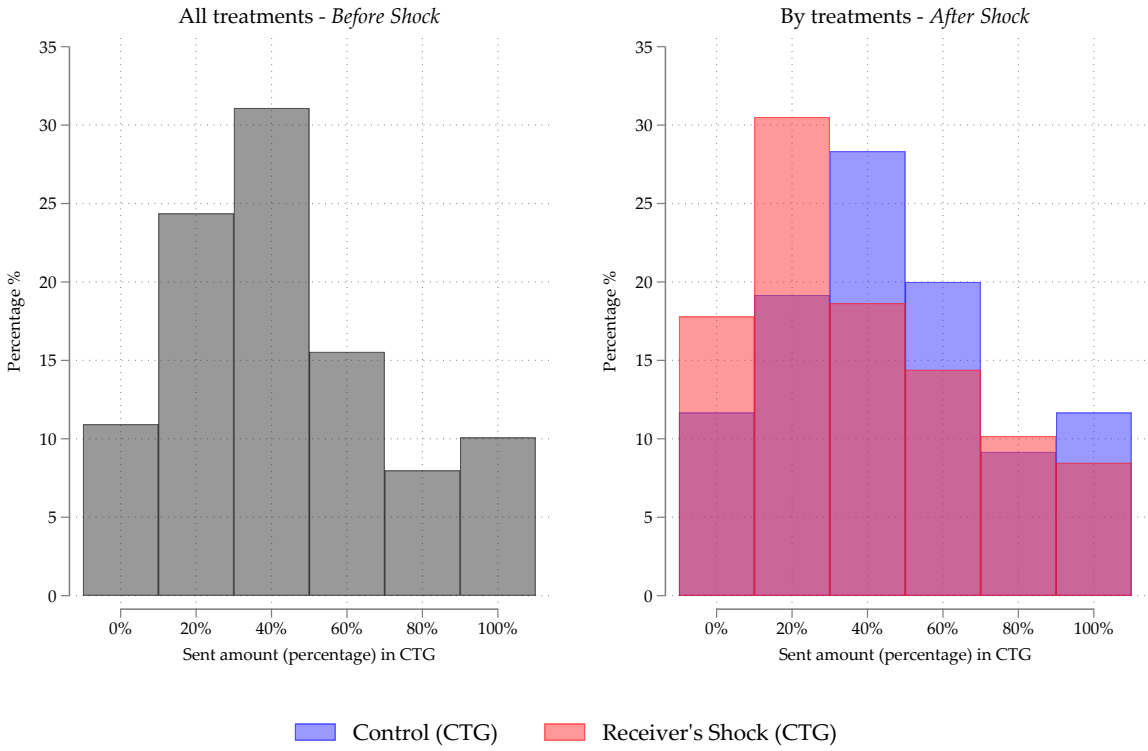
Figure C1: Collective Trust Game design



Note: In red = (*senders*); in blue = (*receivers*); in black = passive agents during each decision. In the first stage, senders transfer from their initial endowment. In the second stage, receivers return from the total transferred. In the next round (**treatment round**), receivers suffer the 60% economic loss. A similar figure showing the new composition of the groups was presented to the experimental subjects.

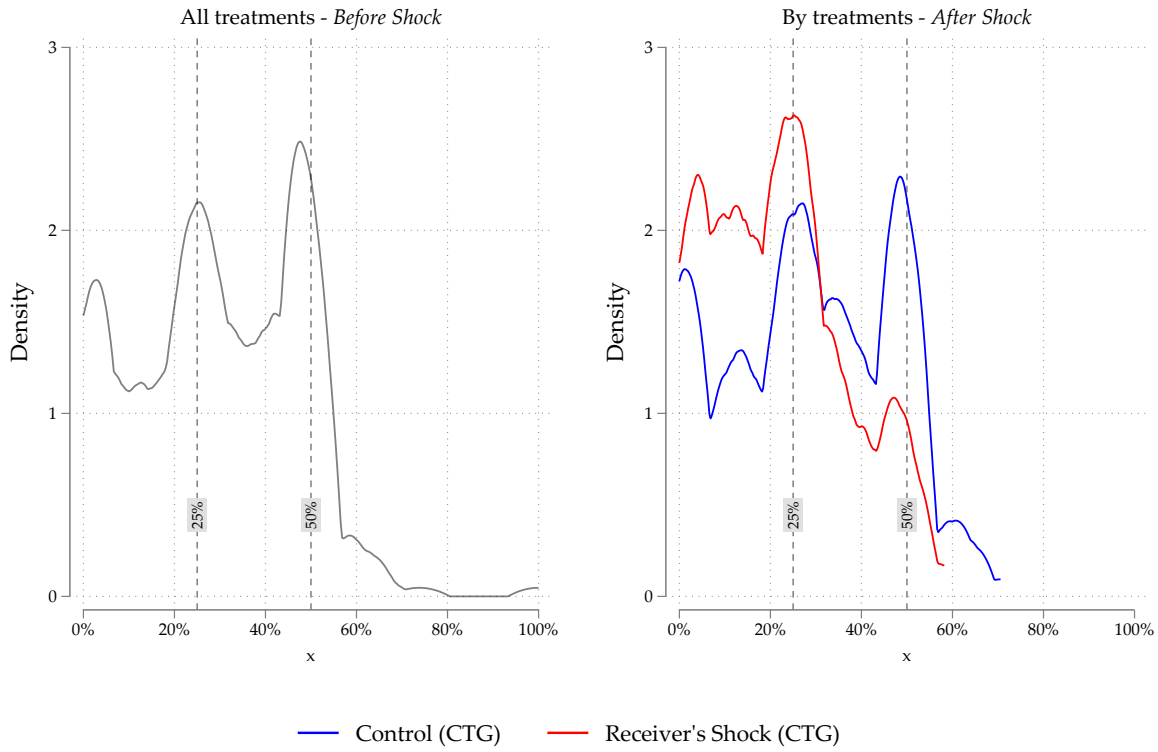
D Additional Figures & Tables

Figure D1: Distribution of Sent amount in Collective Trust Game (CTG)



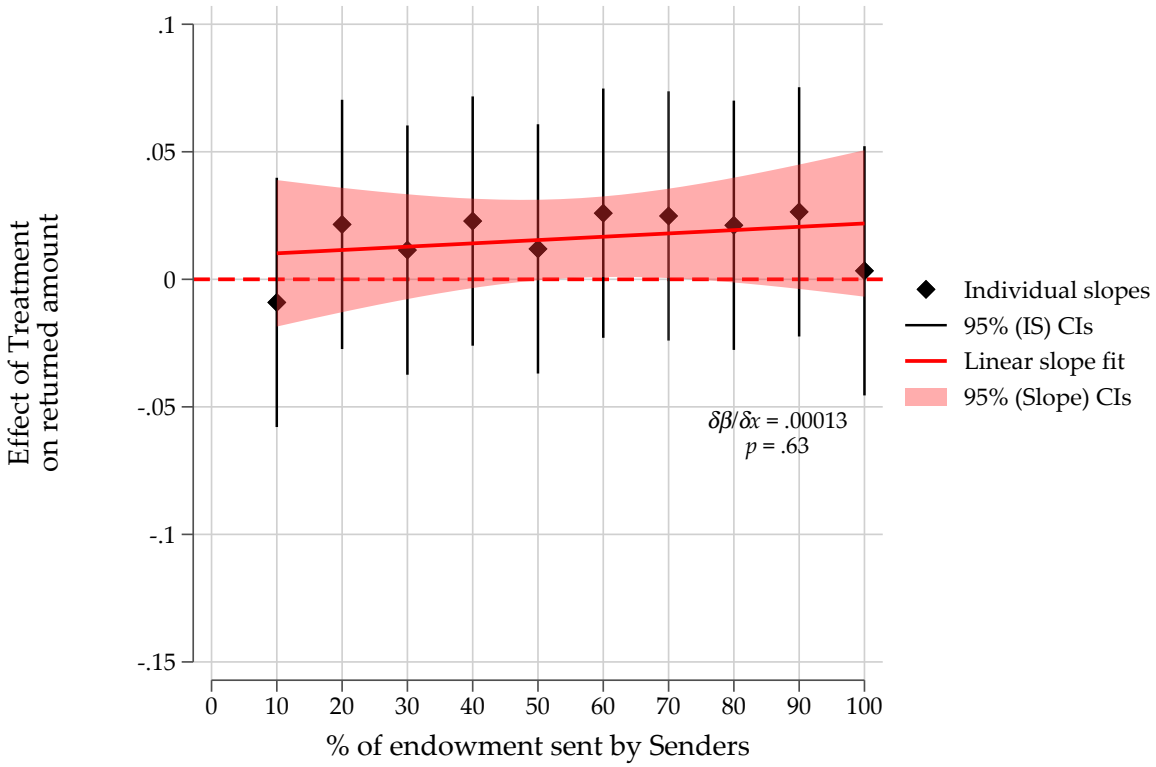
Note: Receiver's shock (**Treatment**). **Treatment**: 60% reduction in the receiver's initial endowment. Left panel (black) presents the distribution (histogram) of the sent amount by senders in the round before the shock. Both treatments are aggregated in this distribution, as there are no differences in their behaviors. Right panel presents the distribution (histogram) of the sent amount by senders in the round after the shock. The distribution of the sent amount for the control group is presented in blue. The distribution of the sent amount for the treatment group is presented in red.

Figure D2: Distribution of the Average returned amount in Collective Trust Game (CTG)



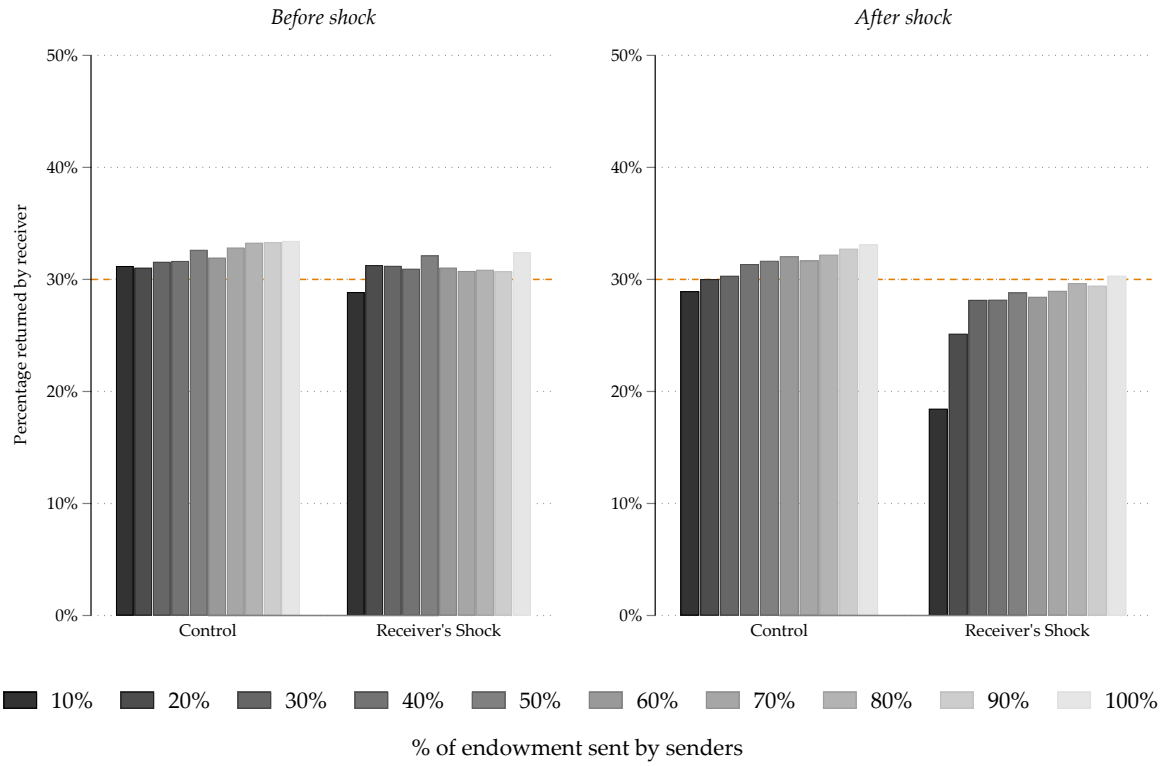
Note: Receiver's shock (**Treatment**). **Treatment**: 60% reduction in the receiver's initial endowment. Left panel (black) presents the distribution (kernel density) of the average returned amount by receivers in the round before the shock. Both treatments are aggregated in this distribution, as there are no differences in their behaviors. Right panel presents the distribution (kernel density) of the average returned amount by receivers in the round after the shock. The distribution of the average returned amount for the control group is presented in blue. The distribution of the average returned amount for the treatment group is presented in red.

Figure D3: **Trustworthiness** across sent amount - (Positive reciprocity).
Round before shock



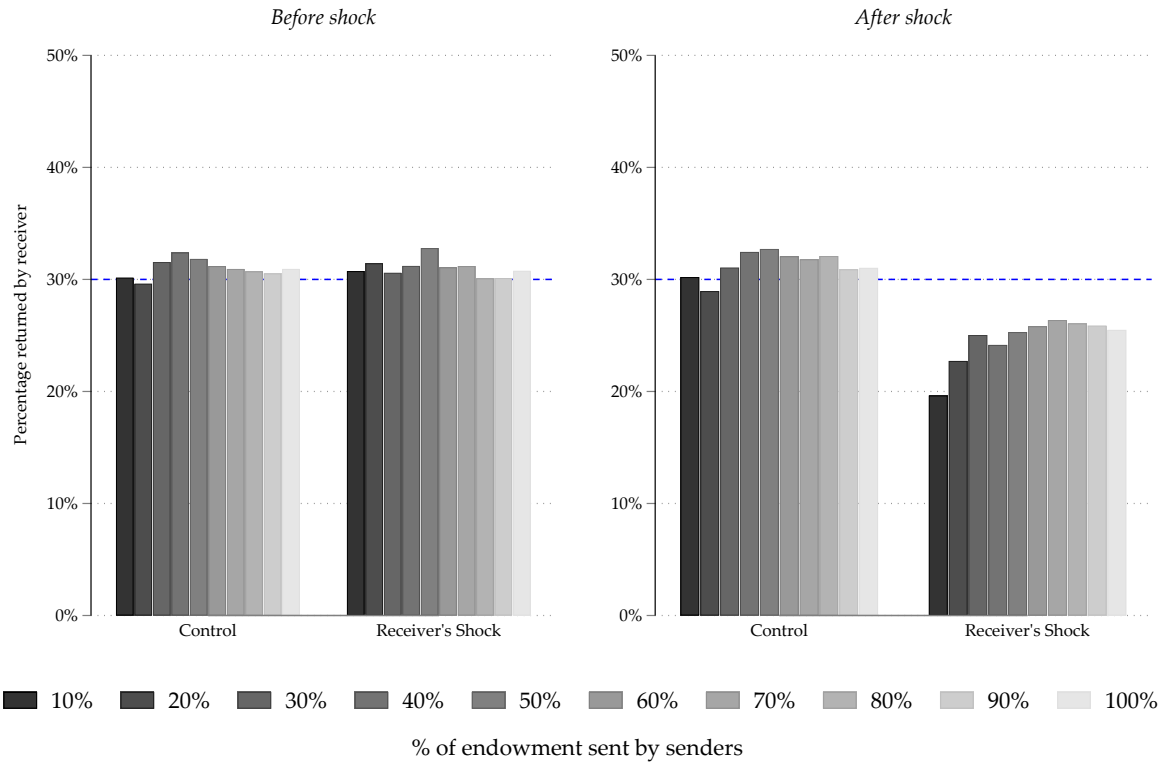
Note: Conditional marginal treatment effect over trustworthiness. Linear trend presented in red. Confidence interval for the linear slope at 95% level. In diamonds, individual treatment effects for each possible transferred amount. 95% confidence intervals for individual slopes (treatment effects) are depicted in black. Finally, $\frac{\partial \beta}{\partial x}$ denote the constant marginal change across transferred amounts. $^{***}p < 0.01,^{**}p < 0.05,^{*}p < 0.1$

Figure D4: Sender beliefs on the returned amount in the Collective Trust Game



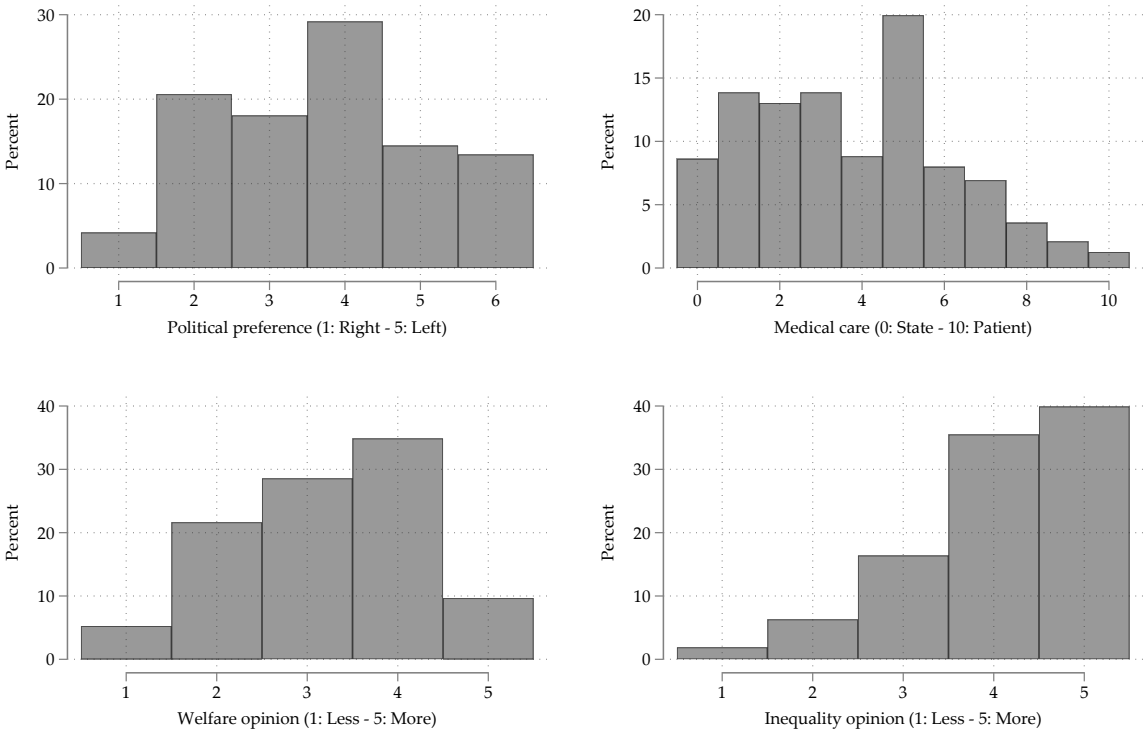
Note: Receiver's shock (**Treatment**). **Treatment**: 60% reduction in the receiver's initial endowment. Sender's initial endowment remain = 100 EP for both treatments and rounds. Receiver's initial endowment = 100 EP for both treatments. Receiver's treatment endowment in the second round = 60 EP. **Left panel:** Denotes the expected returned amount in the round before the shock. **Right panel:** Denotes the expected returned amount in the round after the shock. Given the strategy method, sender respond: How much do you think the group of receivers will return if your group sent ... % of their endowment. Each bar denotes one of the 10 responses. No controls included.

Figure D5: Receiver beliefs on the returned amount in the Collective Trust Game



Note: Receiver's shock (**Treatment**). **Treatment**: 60% reduction in the receiver's initial endowment. Sender's initial endowment remain = 100 EP for both treatments and rounds. Receiver's initial endowment = 100 EP for both treatments. Receiver's treatment endowment in the second round = 60 EP. **Left panel:** Denotes the expected returned amount in the round before the shock. **Right panel:** Denotes the expected returned amount in the round after the shock. Given the strategy method, receiver respond: How much do you think the other receiver will return if senders sent ... % of their endowment. Each bar denotes one of the 10 responses. No controls included.

Figure D6: Distribution of variables (Z_i)



Note: Figure presents the distribution (histogram) of the political preferences and personal attitudes. Both treatments are aggregated in this distribution, as there are no differences in their political preferences and personal attitudes. **Political preference**: denotes the participant's answer to the following question: When you think about your political viewpoint, would you describe yourself as right-wing (1), center-right (2), center (3), center-left (4), left-wing (5), does not know (6)? **Medical care**: is a variable that indicates between 0 (State) and 10 (Patient) who should paid more for a medical treatment. **Welfare opinion**: is a categorical variable that ranges from 1 = "Totally disagree" to 5 = "Totally agree" to the statement: "Each individual is responsible for their own well-being". **Inequality opinion** is a categorical variable that ranges from 1 = "Totally disagree" to 5 = "Totally agree" to the statement: "The government should implement strong policies to reduce inequality".

Figure D7: Social Norm - **Trust**

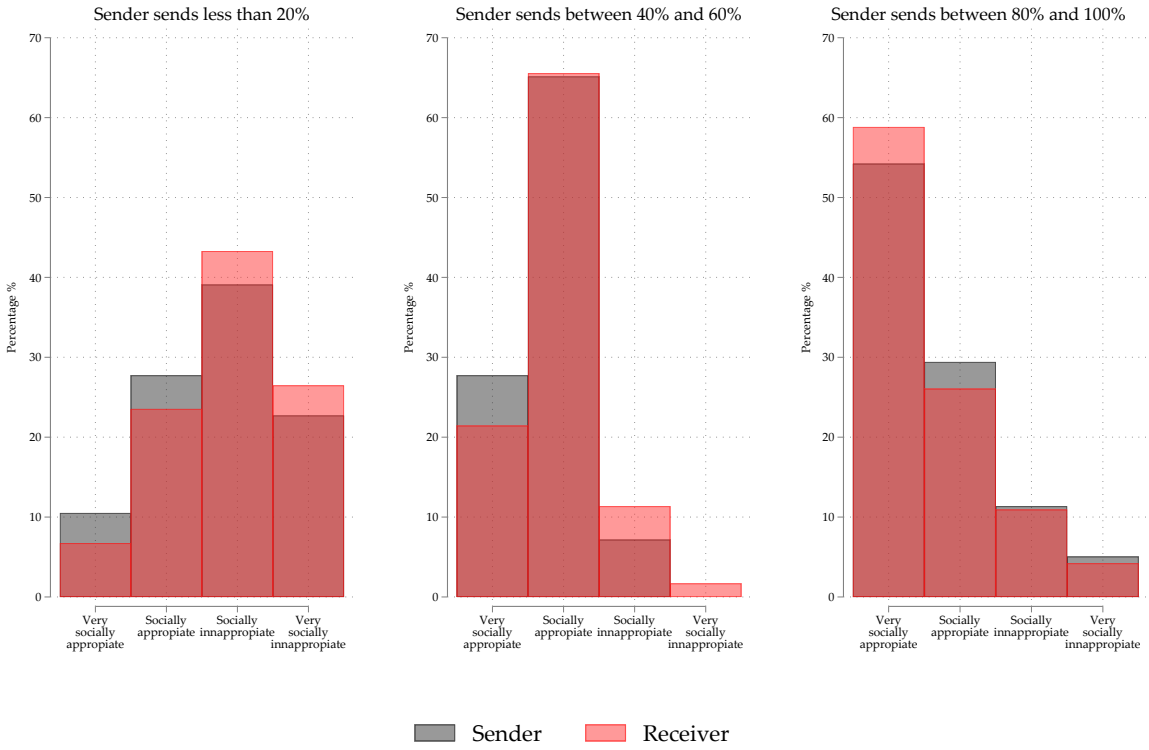


Figure D8: Social Norm - **Trustworthiness**

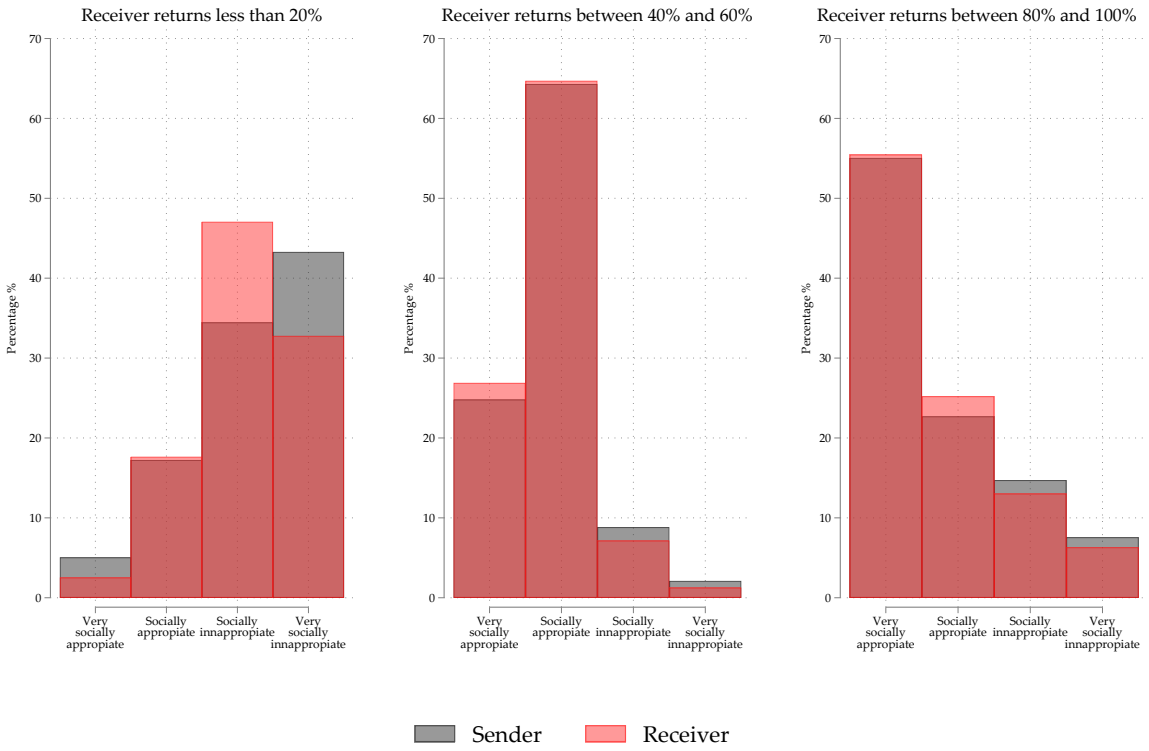


Table D1: Collective **Trustworthiness** - Positive Reciprocity

Dep Var: **Average amount returned**	Round			
	Before shock		After shock	
	(1)	(2)	(3)	(4)
Treatment	-0.010	-0.011	-0.097***	-0.100***
	(0.03)	(0.03)	(0.03)	(0.03)
Amount sent = 20	-0.015*	-0.015*	-0.003	-0.003
	(0.01)	(0.01)	(0.01)	(0.01)
Amount sent = 30	0.010	0.010	0.011	0.011
	(0.01)	(0.01)	(0.01)	(0.01)
Amount sent = 40	0.000	0.000	0.023**	0.023**
	(0.01)	(0.01)	(0.01)	(0.01)
Amount sent = 50	0.015	0.015	0.026*	0.026*
	(0.02)	(0.02)	(0.01)	(0.01)
Amount sent = 60	-0.001	-0.001	0.018	0.018
	(0.02)	(0.02)	(0.02)	(0.02)
Amount sent = 70	-0.004	-0.004	0.019	0.019
	(0.02)	(0.02)	(0.02)	(0.02)
Amount sent = 80	-0.005	-0.005	0.004	0.004
	(0.02)	(0.02)	(0.02)	(0.02)
Amount sent = 90	-0.010	-0.010	0.001	0.001
	(0.02)	(0.02)	(0.02)	(0.02)
Amount sent = 100	0.011	0.011	0.013	0.013
	(0.02)	(0.02)	(0.02)	(0.02)
Treatment × Amount sent = 20	0.031**	0.031**	0.015	0.015
	(0.01)	(0.01)	(0.01)	(0.01)
Treatment × Amount sent = 30	0.021	0.021	0.015	0.015
	(0.02)	(0.02)	(0.02)	(0.02)
Treatment × Amount sent = 40	0.032*	0.032*	0.014	0.014
	(0.02)	(0.02)	(0.02)	(0.02)
Treatment × Amount sent = 50	0.021	0.021	0.016	0.016
	(0.02)	(0.02)	(0.02)	(0.02)
Treatment × Amount sent = 60	0.035	0.035	0.031	0.031
	(0.02)	(0.02)	(0.02)	(0.02)
Treatment × Amount sent = 70	0.034	0.034	0.039*	0.039*
	(0.02)	(0.02)	(0.02)	(0.02)
Treatment × Amount sent = 80	0.030	0.030	0.045*	0.045*
	(0.02)	(0.02)	(0.02)	(0.02)
Treatment × Amount sent = 90	0.035	0.035	0.048**	0.048**
	(0.02)	(0.02)	(0.02)	(0.02)
Treatment × Amount sent = 100	0.012	0.012	0.043*	0.043*
	(0.02)	(0.02)	(0.02)	(0.02)
Constant	0.284***	0.011	0.274***	0.005
	(0.02)	(0.10)	(0.02)	(0.10)
Mean Dep. Variable	0.3	0.3	0.3	0.3
Effect size (%)	-3.6	-3.8	-34.1	-35.0
Sociodemographic controls		✓		✓
Observations	2380	2380	2380	2380
Clusters	238	238	238	238
Adjusted R-squared	0.00	0.07	0.04	0.11

Note: The set of controls included in ***Controls*** are: age, gender, stratum, risk measure, the sent amount in regular trust game, previous experience participating in experiments, and medical care -as it shows unbalance at 10% level (p-value = 0.06)-. The set of controls included in ***Beliefs*** controls are the expected returned amount by the sender for each possible transferred amount (strategy method). Standard errors are clustered at the participant level. $^{***}p < 0.01$,** $p < 0.05$,* $p < 0.1$

Table D2: Beliefs of **Trustworthiness** - Expected Positive Reciprocity

Dep Var: **Receiver beliefs on the returned amount by his partner**	Round			
	Before shock		After shock	
	(1)	(2)	(3)	(4)
Treatment	0.006	0.003	-0.106***	-0.110***
	(0.03)	(0.03)	(0.03)	(0.02)
Amount sent = 20	-0.005	-0.005	-0.012	-0.012
	(0.01)	(0.01)	(0.01)	(0.01)
Amount sent = 30	0.014	0.014	0.009	0.009
	(0.01)	(0.01)	(0.01)	(0.01)
Amount sent = 40	0.022*	0.022*	0.022	0.022
	(0.01)	(0.01)	(0.01)	(0.01)
Amount sent = 50	0.017	0.017	0.025*	0.025*
	(0.02)	(0.02)	(0.02)	(0.02)
Amount sent = 60	0.010	0.010	0.019	0.019
	(0.02)	(0.02)	(0.01)	(0.01)
Amount sent = 70	0.008	0.008	0.016	0.016
	(0.02)	(0.02)	(0.01)	(0.01)
Amount sent = 80	0.006	0.006	0.019	0.019
	(0.02)	(0.02)	(0.02)	(0.02)
Amount sent = 90	0.004	0.004	0.007	0.007
	(0.02)	(0.02)	(0.02)	(0.02)
Amount sent = 100	0.008	0.008	0.008	0.008
	(0.02)	(0.02)	(0.02)	(0.02)
Treatment × Amount sent = 20	0.012	0.012	0.043***	0.043***
	(0.02)	(0.02)	(0.01)	(0.01)
Treatment × Amount sent = 30	-0.015	-0.015	0.045**	0.045**
	(0.02)	(0.02)	(0.02)	(0.02)
Treatment × Amount sent = 40	-0.018	-0.018	0.023	0.023
	(0.02)	(0.02)	(0.02)	(0.02)
Treatment × Amount sent = 50	0.004	0.004	0.031	0.031
	(0.02)	(0.02)	(0.02)	(0.02)
Treatment × Amount sent = 60	-0.007	-0.007	0.043**	0.043**
	(0.02)	(0.02)	(0.02)	(0.02)
Treatment × Amount sent = 70	-0.003	-0.003	0.052**	0.052**
	(0.02)	(0.02)	(0.02)	(0.02)
Treatment × Amount sent = 80	-0.012	-0.012	0.046**	0.046**
	(0.02)	(0.02)	(0.02)	(0.02)
Treatment × Amount sent = 90	-0.010	-0.010	0.056**	0.056**
	(0.02)	(0.02)	(0.02)	(0.02)
Treatment × Amount sent = 100	-0.007	-0.007	0.050**	0.050**
	(0.02)	(0.02)	(0.02)	(0.02)
Constant	0.302***	0.071	0.302***	0.128
	(0.02)	(0.10)	(0.02)	(0.09)
Mean Dep. Variable	0.3	0.3	0.3	0.3
Effect size (%)	1.9	0.9	-33.7	-35.2
Sociodemographic controls		✓		✓
Observations	2380	2380	2380	2380
Clusters	238	238	238	238
Adjusted R-squared	0.00	0.08	0.04	0.13

Note: The set of controls included in ***Controls*** are: age, gender, stratum, risk measure, the sent amount in regular trust game, previous experience participating in experiments, and medical care -as it shows unbalance at 10% level (p-value = 0.06)-. The set of controls included in ***Beliefs*** controls are the expected returned amount by the sender for each possible transferred amount (strategy method). Standard errors are clustered at the participant level. $^{***}p < 0.01, ^{**}p < 0.05, ^{*}p < 0.1$

Table D3: Heterogeneous treatment effects over **Collective Trust**

Dep Var: **Sent Amount**	Political preference or attitude			
	Political preference	Medical Care	Welfare Opinion	Inequality opinion
	(1)	(2)	(3)	(4)
Treatment	-13.39	-27.64***	-21.30*	17.90
	(10.45)	(6.80)	(10.90)	(15.75)
Z_i variable	2.11	-4.23***	0.71	3.98
	(2.03)	(0.88)	(2.50)	(2.64)
Treatment $\times$ Z_i variable	1.39	5.09***	4.43	-6.26
	(2.72)	(1.47)	(3.27)	(3.79)
Constant	38.59***	63.34***	43.84***	30.43***
	(7.68)	(4.71)	(8.50)	(10.80)
Observations	238	238	238	238
Adjusted R-squared	0.0	0.1	0.0	0.0

Note: Z_i denotes the political preference or attitude. **Political preference**: denotes the participant's answer to the following question: When you think about your political viewpoint, would you describe yourself as right-wing (1), center-right (2), center (3), center-left (4), left-wing (5), does not know (6)? **Medical care**: is a variable that indicates between 0 (State) and 10 (Patient) who should paid more for a medical treatment. **Welfare opinion**: is a categorical variable that ranges from 1 = "Totally disagree" to 5 = "Totally agree" to the statement: "Each individual is responsible for their own well-being". **Inequality opinion** is a categorical variable that ranges from 1 = "Totally disagree" to 5 = "Totally agree" to the statement: "The government should implement strong policies to reduce inequality". Robust standard errors in parenthesis. $^{***}p < 0.01$,** $p < 0.05$,* $p < 0.1$

Table D4: Heterogeneous treatment effects over **Collective Trustworthiness**

Dep Var: **Average returned amount**	Political preference or attitude			
	Political preference	Medical Care	Welfare Opinion	Inequality opinion
	(1)	(2)	(3)	(4)
Treatment	-0.08	-0.00	0.04	-0.07
	(0.06)	(0.04)	(0.07)	(0.08)
Z_i variable	0.02	0.00	0.02	0.01
	(0.01)	(0.01)	(0.02)	(0.02)
Treatment $\times$ Z_i variable	0.00	-0.02**	-0.04*	0.00
	(0.02)	(0.01)	(0.02)	(0.02)
Constant	0.22***	0.27***	0.22***	0.23***
	(0.05)	(0.03)	(0.05)	(0.06)
Observations	238	238	238	238
Adjusted R-squared	0.1	0.1	0.1	0.0

Note: Z_i denotes the political preference or attitude. **Political preference**: denotes the participant's answer to the following question: When you think about your political viewpoint, would you describe yourself as right-wing (1), center-right (2), center (3), center-left (4), left-wing (5), does not know (6)? **Medical care**: is a variable that indicates between 0 (State) and 10 (Patient) who should paid more for a medical treatment. **Welfare opinion**: is a categorical variable that ranges from 1 = "Totally disagree" to 5 = "Totally agree" to the statement: "Each individual is responsible for their own well-being". **Inequality opinion** is a categorical variable that ranges from 1 = "Totally disagree" to 5 = "Totally agree" to the statement: "The government should implement strong policies to reduce inequality". Robust standard errors in parenthesis. $^{***}p < 0.01$,** $p < 0.05$,* $p < 0.1$

D.1 Robustness Checks

While randomization allows me to compare each group within the round, it is possible to estimate the treatment effect employing a Two-Way Fixed Effect (TWFE) model comparing both rounds and groups. Therefore, I present this estimation as a robustness check for both estimates: Collective

Trust and Collective Trustworthiness.

Collective Trust

I propose the following alternative model to estimate the treatment effect over collective trust,

$$Sent_{r,g} = \phi_g + \eta_r + \tau D_{g,r} + \varepsilon_{g,r}, \tag{21}$$

where ϕ_g denotes the fixed effect for each group g (control and treated group), η_r denotes the fixed effect for each round r (before and after the shock). Finally, $D_{g,r}$ is dummy variable that takes the value of = 1 if the group is treated in the second round, 0 otherwise. The OLS estimates for Equation (21) are presented in Table D5.

Table D5: Collective **Trust** - TWFE Estimates

Dep Var: **Sent amount**	TWFE	
	(1)	(2)
Treatment	-0.45	0.43
	(3.72)	(3.53)
Round	2.83	2.83
	(2.03)	(2.04)
Treatment × Round	-6.90**	-6.90**
	(2.95)	(2.96)
Constant	43.33***	37.17**
	(2.67)	(15.11)
Mean Dep. Variable	43.3	43.3
Effect size (%)	-15.9	-15.9
Sociodemographic controls		✓
Observations	476	476
Clusters	238	238
Adjusted R-squared	0.01	0.11

Note: The set of controls included in ***Controls*** are: age, gender, stratum, risk measure, previous experience participating in experiments, and medical care -as it shows unbalance at 10% level (p-value = 0.06)-. Clustered standard errors at participant level in parenthesis. $^{***}p < 0.01$,$^{**}p < 0.05$,$^{*}p < 0.1$

Collective Trustworthiness

I propose the following alternative model to estimate the treatment effect over collective trustworthiness,

$$AverageReturned_{r,g} = \phi_g + \eta_r + \tau D_{g,r} + \varepsilon_{g,r}, \tag{22}$$

where ϕ_g denotes the fixed effect for each group g (control and treated group), η_r denotes the fixed effect for each round r (before and after the shock). Finally, $D_{g,r}$ is dummy variable that takes the value of = 1 if the group is treated in the second round, 0 otherwise. The OLS estimates for Equation (22) are presented in Table D6.

Table D6: Collective **Trustworthiness** - TWFE Estimates

Dep Var: **Average returned amount**	TWFE	
	(1)	(2)
Treatment	0.01	0.01
	(0.02)	(0.02)
Round	0.00	0.00
	(0.01)	(0.01)
Treatment × Round	-0.09***	-0.09***
	(0.01)	(0.01)
Constant	0.28***	0.01
	(0.02)	(0.10)
Mean Dep. Variable	0.3	0.3
Effect size (%)	-30.0	-30.0
Sociodemographic controls		✓
Observations	476	476
Clusters	238	238
Adjusted R-squared	0.03	0.11

Note: The set of controls included in ***Controls*** are: age, gender, stratum, risk measure, previous experience participating in experiments, and medical care -as it shows unbalance at 10% level (p-value = 0.06)-. The set of controls included in ***Beliefs*** controls are the expected returned amount by the sender for each possible transferred amount (strategy method). Clustered standard errors at participant level in parenthesis. $^{***}p < 0.01$,$^{**}p < 0.05$,$^{*}p < 0.1$

E Experimental Screen Instructions

Original instructions in Spanish

Figure E1: General instructions

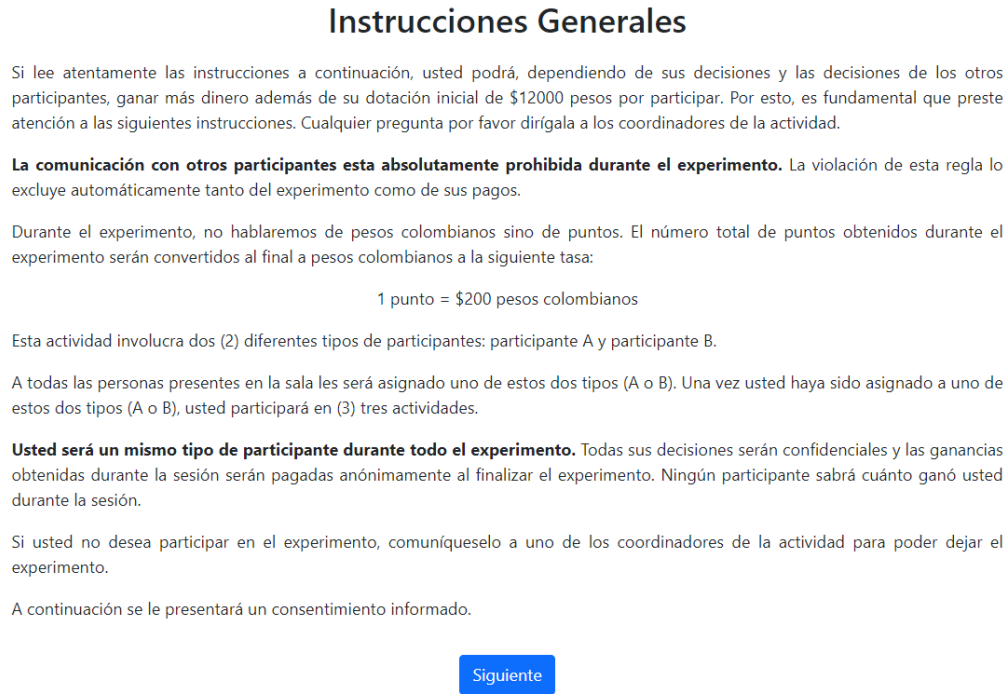


Figure E2: General Instructions - 2



Figure E3: Welcome - Roles during the experiment

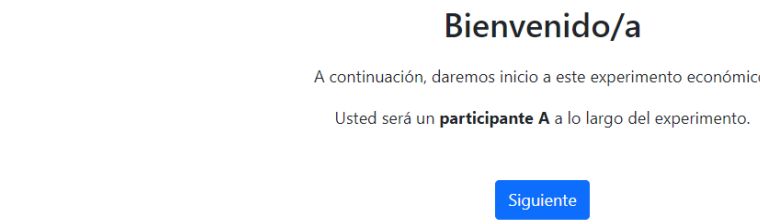


Figure E4: Trust Game Instructions - 1

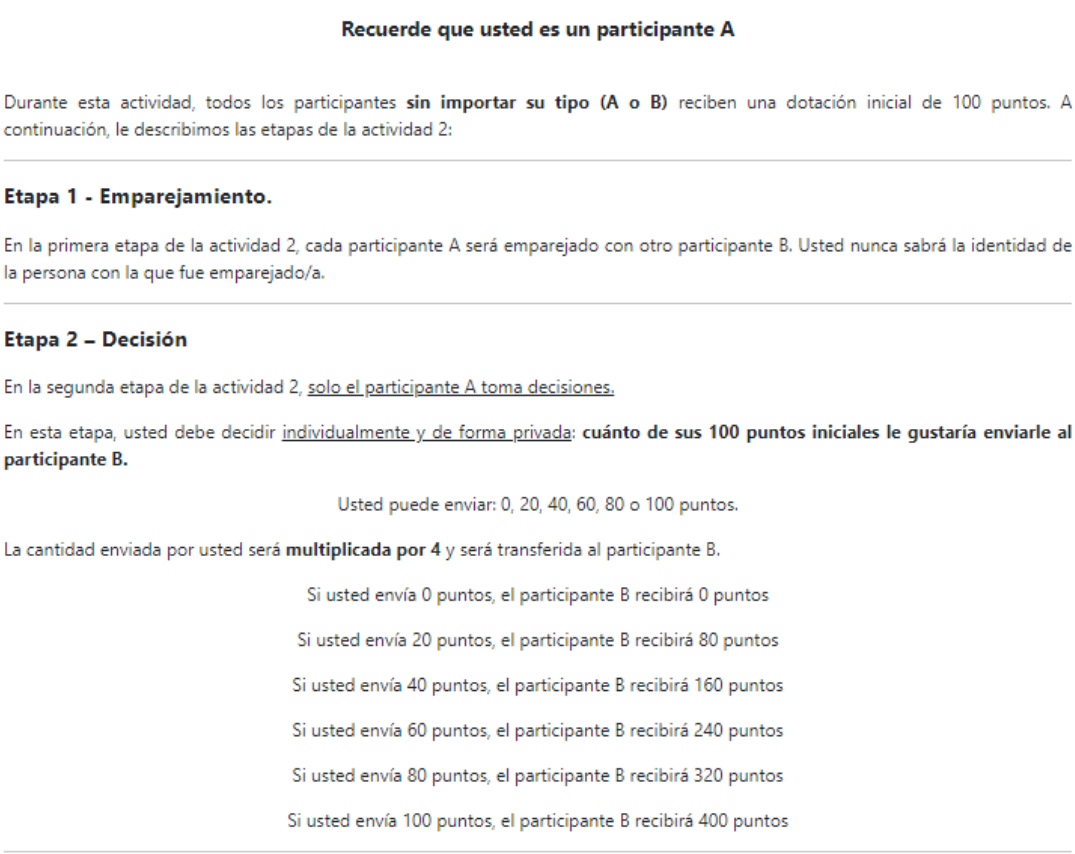


Figure E5: Trust Game Instructions - 2

Etapa 3 – Devolución

En la tercera etapa de la actividad 2, solo el participante B toma decisiones.

En esta tercera etapa, el participante B deberá decidir individualmente y de forma privada: **cuánto del total de puntos recibidos quiere devolverle a usted.**

El participante B puede devolver al participante A un número de puntos entre 0 y el total de puntos recibidos.

Los puntos que el participante B le devuelve al participante A **no se multiplican.** Esto significa que usted recibirá la misma cantidad de puntos que el participante B decida devolverle.

Finalmente, note que el participante B no puede usar puntos de su dotación inicial para devolverle al participante A.

A continuación le presentamos una representación gráfica de la actividad 2.

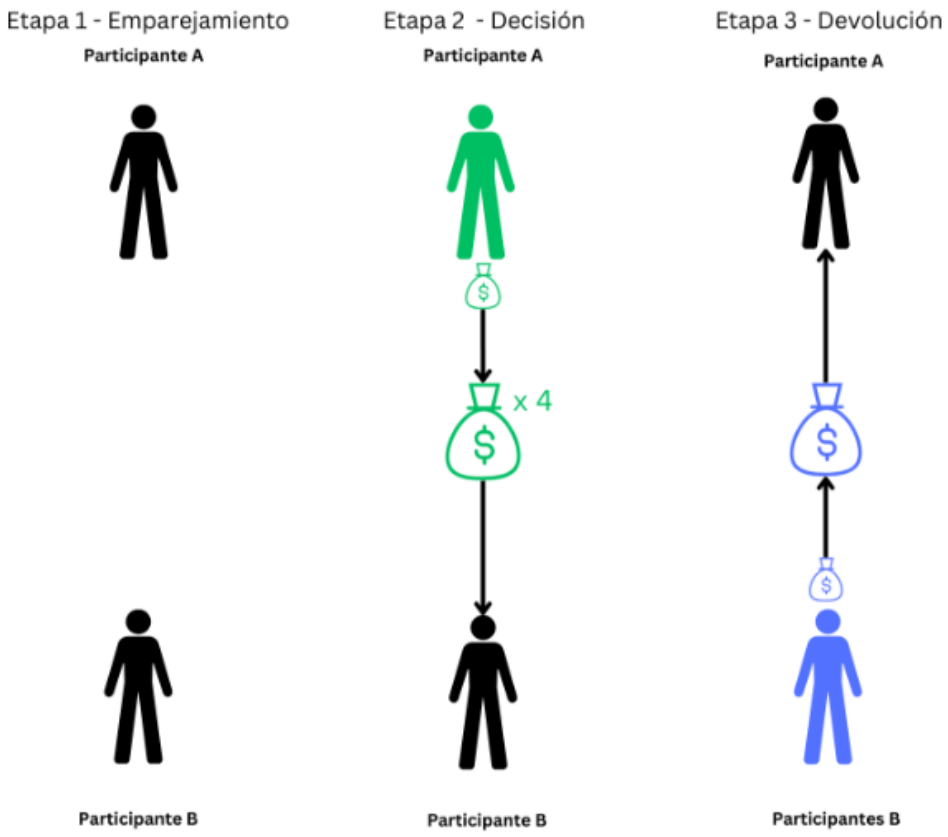


Figure E6: Trust Game - Control Question

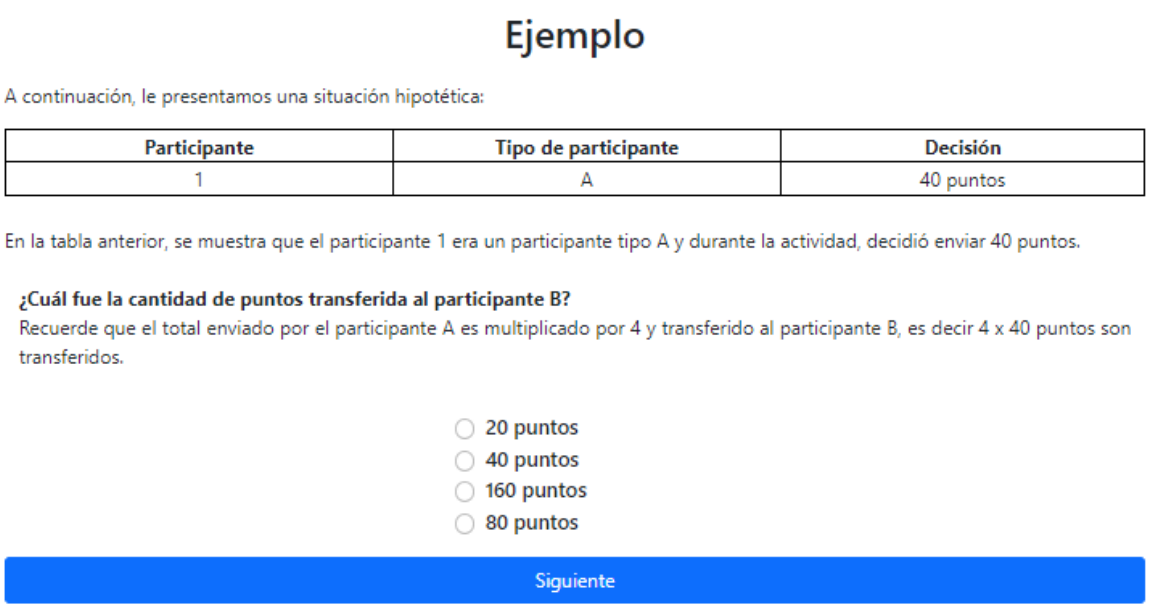


Figure E7: Trust Game - Decision

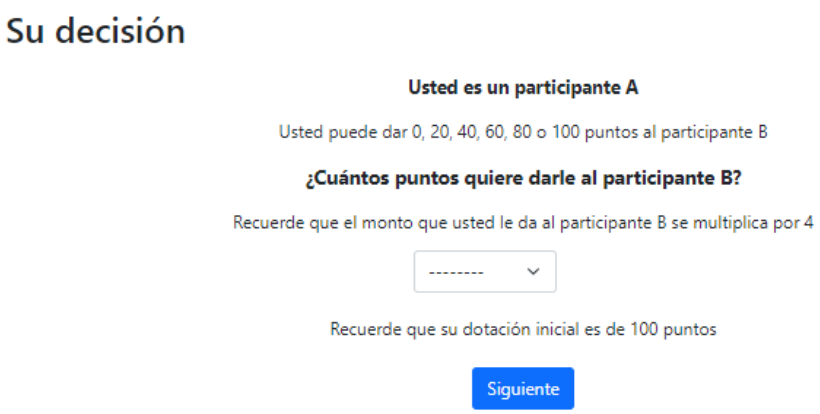


Figure E8: Trust Game - Beliefs



Figure E9: Collective Trust Game - Welcome

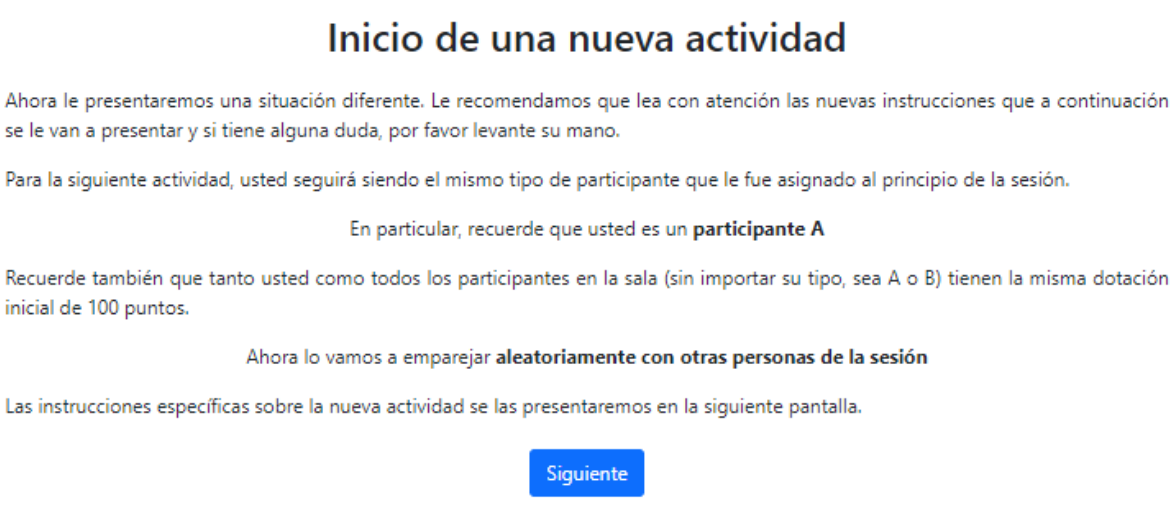


Figure E10: Collective Trust Game Instructions - 1

Durante esta actividad, todos los participantes **sin importar su tipo (A o B)** reciben una dotación inicial de 100 puntos. A continuación, le describimos las etapas de la actividad 3:

Etapa 1 - Emparejamiento.

En la primera etapa de la actividad 3, los participantes A serán agrupados en grupos de dos (2) y los participantes B serán agrupados en grupos de dos (2).

Posteriormente, cada grupo de participantes A será emparejado con un grupo de participantes B formando un grupo final de **cuatro (4)** participantes por grupo. Usted nunca sabrá la identidad de las personas con las que fue emparejado/a.

Etapa 2 – Decisión

En la segunda etapa de la actividad 3, solo los participantes A toman decisiones.

En esta etapa, usted debe decidir individualmente y de forma privada: **cuánto de sus 100 puntos iniciales le gustaría enviarle al grupo de participantes B.** Usted puede enviarles a los participantes B de su grupo: 0, 20, 40, 60, 80 o 100 puntos.

El otro participante A de su grupo también decidirá individualmente y de forma privada, cuánto de sus 100 puntos iniciales quiere enviarles a los participantes B.

Posteriormente, la cantidad enviada por usted será sumada con la cantidad enviada por el otro participante A de su grupo, el total de la suma será **multiplicado por 4** y será transferida en partes iguales a los participantes B de su grupo.

Si su grupo envía 0 puntos, el grupo de participantes B recibirá 0 puntos

Si su grupo envía 20 puntos, el grupo de participantes B recibirá 80 puntos

Si su grupo envía 40 puntos, el grupo de participantes B recibirá 160 puntos

Si su grupo envía 60 puntos, el grupo de participantes B recibirá 240 puntos

Si su grupo envía 80 puntos, el grupo de participantes B recibirá 320 puntos

Si su grupo envía 100 puntos, el grupo de participantes B recibirá 400 puntos

Si su grupo envía 120 puntos, el grupo de participantes B recibirá 480 puntos

Si su grupo envía 140 puntos, el grupo de participantes B recibirá 560 puntos

Si su grupo envía 160 puntos, el grupo de participantes B recibirá 640 puntos

Si su grupo envía 180 puntos, el grupo de participantes B recibirá 720 puntos

Si su grupo envía 200 puntos, el grupo de participantes B recibirá 800 puntos

Recuerde que el total recibido por el grupo de participantes B se reparte en **partes iguales** entre ambos participantes B.

Figure E11: Collective Trust Game Instructions - 2

Etapa 3 – Devolución

En la tercera etapa de la actividad 3, solo los participantes B toman decisiones.

Una vez cada participante B haya recibido la cantidad transferida que le corresponde, cada participante B deberá decidir: **cuánto del total de sus puntos recibidos quiere devolver a los participantes A del grupo.**

Recuerde que cada participante B recibe la mitad de la cantidad transferida al grupo de participantes B.

Cada participante B decidirá individualmente y de forma privada, cuánto devolver a los participantes A del grupo.

Cada participante B puede escoger devolver cualquier número de puntos entre 0 y el total de puntos recibidos.

Posteriormente, la cantidad de vuelta por cada participante B será sumada y transferida (sin multiplicar) a los participantes A del grupo en partes iguales.

A continuación le presentamos una representación gráfica de la actividad 3.

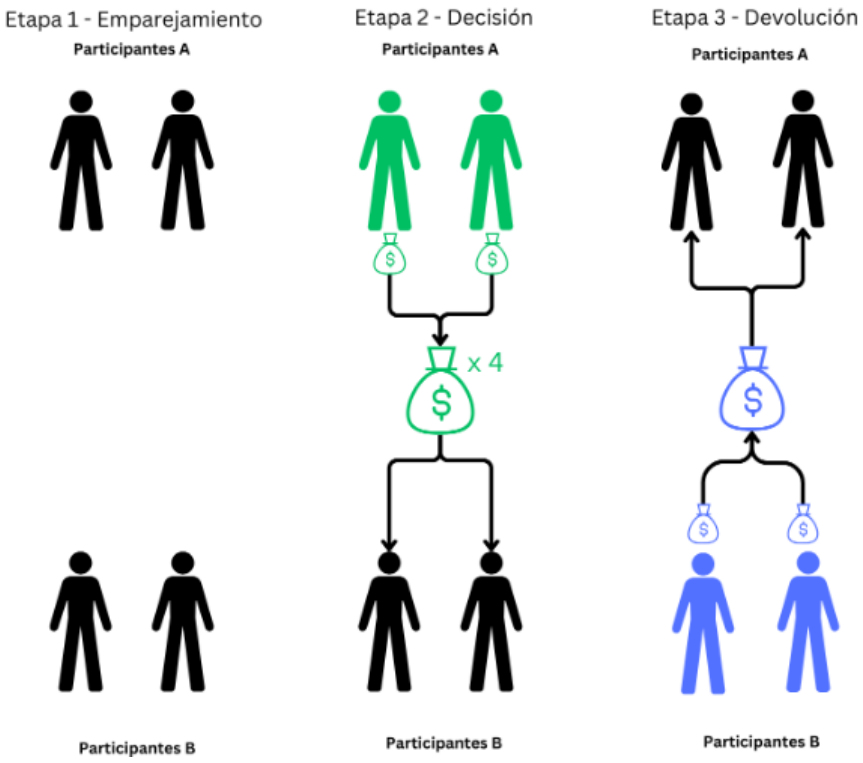


Figure E12: Collective Trust Game - Control Question

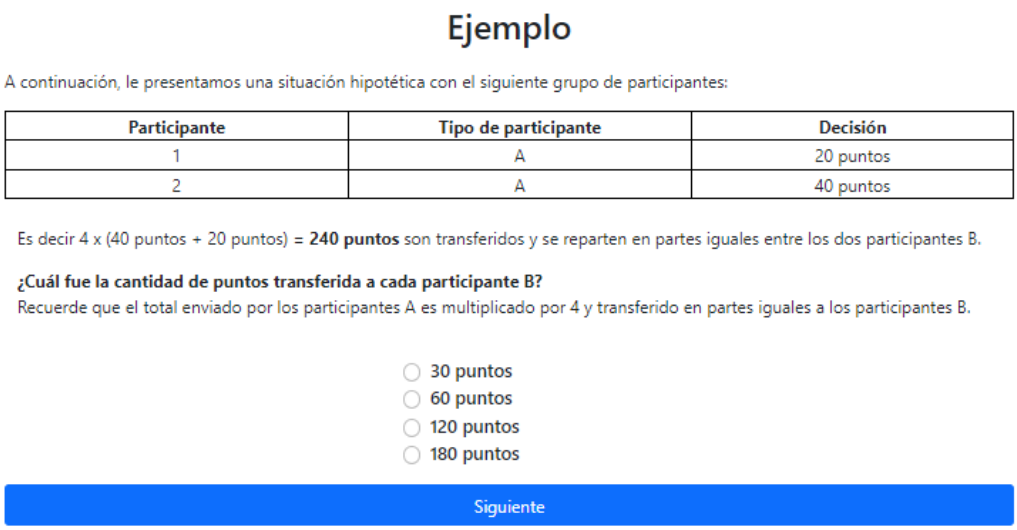


Figure E13: Collective Trust Game - Control Question 2

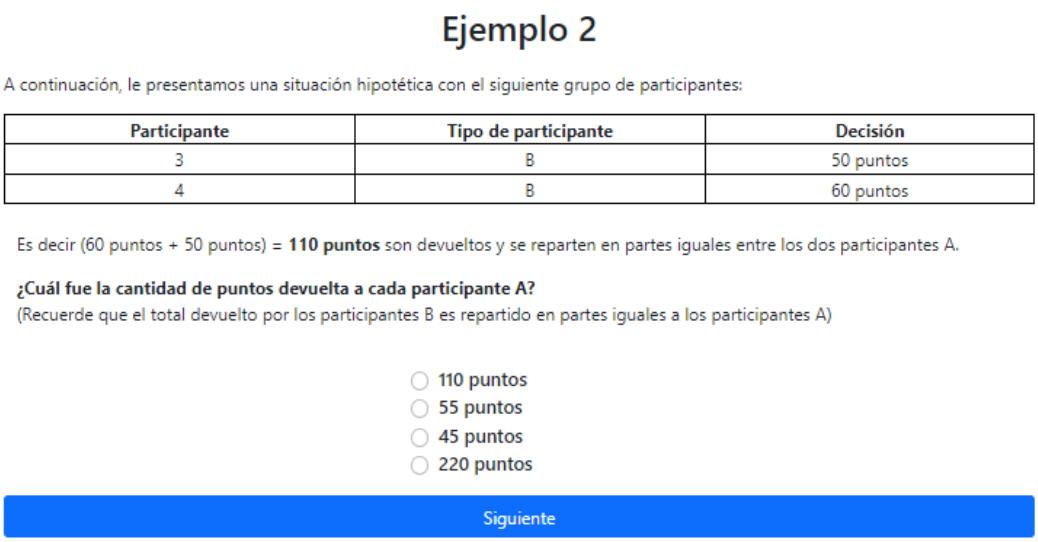


Figure E14: Collective Trust Game - Decision

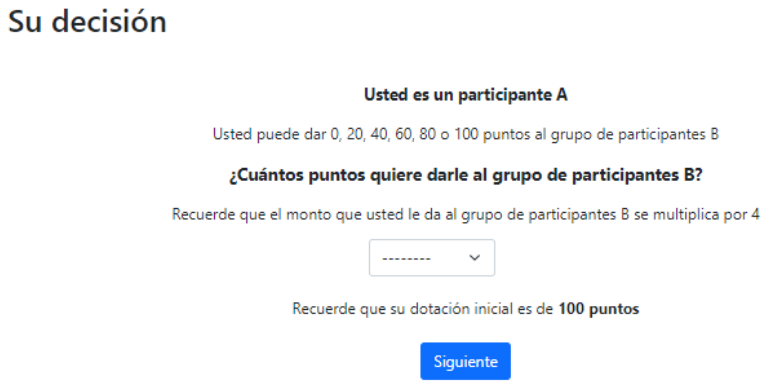


Figure E15: Collective Trust Game - Beliefs

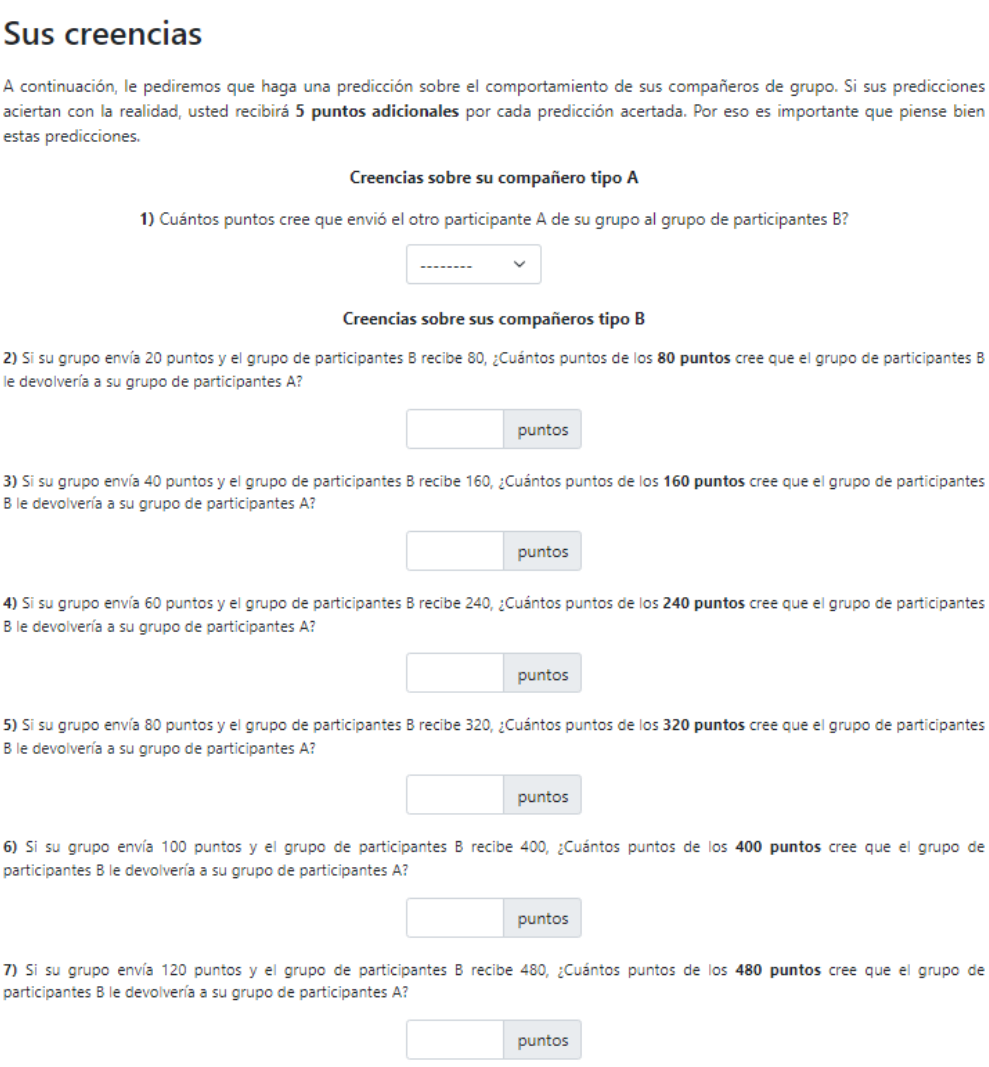


Figure E16: Announcement - Negative Economic Shock

Choque negativo

Por favor, lea atentamente las siguientes instrucciones



Nueva situación:

Acaba de ocurrir un imprevisto y ahora los participantes B se verán afectados por un choque negativo que destruyó su dotación inicial en un **60%**.

Por este motivo, la dotación inicial de cada participante B dejará de ser 100 puntos y ahora será solo de **40 puntos**.

Recuerde que usted es un participante A

Ni usted ni ninguno de los participantes A se vió afectado por el choque. Por este motivo, usted y todos los participantes A sí mantendrán su dotación inicial de 100 puntos.